What a Faithfulness Score Can See: Injected Errors, Answer Length and Resolving Power in Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation is often guarded by a faithfulness score, which decides whether the retrieved documents support an answer, and the same scores are used to compare systems. Such scores are ranked by how well they agree with human hallucination labels, or tested on reference summaries edited to contain one error. Neither shows whether a scorer registers the same error equally in answers from different generators, or how small a difference between systems it can resolve.

We hold 32,000 answers fixed — four embedding models, two question answering datasets, four generators — and score every answer with three evaluators. Into up to 200 answers per system, dataset and generator we inject a known error, replacing one grounded value with a value absent from the retrieved context. Under a standard entailment scorer, a fixed threshold catches 5.1% of these edits on the most verbose generator and 56.6% on the tersest: one wrong value is largely diluted among the correct claims around it. Making one model more verbose by prompt alone reproduces the drop, under all three evaluators. A generative fact-checker run on the same cases detects far more, 81 to 98%, and narrows the gap between generators to about 15 points, but the same prompt still costs it 31 and 40 points. Length is not the whole story, since at a matched single assertion the four generators still respond by between 0.102 and 0.485, and removing an opening "according to the provided context" clause from the answer closes part of that gap. On 5,593 human-annotated answers from RAGTruth, every scorer catches a hallucination less often the smaller the share of the answer it occupies.

The evaluators also differ in resolving power. Comparing the four retrieval systems, whether the comparison returns a difference or a null depends on which evaluator scored it in three of eight dataset-generator cells, and in four when the fact-checker is added as a fourth evaluator, although the evaluators are not shown to rank the systems differently. Averaging in refusals, or choosing a different equivalence margin, moves the conclusion as well. A faithfulness threshold should be tested on deliberately falsified answers from the generator it guards, and a faithfulness result should name its evaluator.

1 Introduction

A document question answering system is given a set of uploaded files and must answer from them. If an invoice records a project budget of $1,000, an answer stating $1,500 fails in a specific way: fluent, on topic, citing the right document, wrong in the one place that matters. Such systems are typically guarded by a faithfulness score, which measures whether the retrieved text supports the answer (Maynez et al., 2020). The same scores are used to compare systems (Es et al., 2024; Tamber et al., 2025), so they decide both what a user sees and what a paper concludes.

Many such scores ask a natural language inference model whether the retrieved context entails the answer, taking the maximum over retrieved chunks. Others use a trained alignment model (AlignScore; Zha et al., 2023) or a language model as judge (Saad-Falcon et al., 2024; Tamber et al., 2025). They are ranked by how well they agree with human hallucination labels (Honovich et al., 2022; Tang et al., 2023; Niu et al., 2024; Bao et al., 2025), or tested on reference summaries edited to contain one error (Ma et al., 2023). Neither tells us whether a scorer can separate the systems someone is actually comparing, and pairs built from reference text cannot show whether a scorer treats answers from different generators alike.

This paper asks what such a score can see. We do not propose a new one. We take 32,000 answers from a fixed grid — four embedding models, two datasets (Kwiatkowski et al., 2019; Yang

et al., 2018), four generators — and score every answer with three evaluators. We ask two questions: whether a known error registers equally in answers from different generators, and how small a difference between retrieval systems each evaluator can resolve. The grid is a testbed rather than the object of study: its retrieval systems span 4.4 and 11.9 NDCG@5 points on the two datasets while generation stays fixed. We expected that a score rating a whole answer at once would react less to one wrong value when more claims surround it, so its sensitivity would depend on the generator.

Our main findings are:

- Swapping one grounded number in an answer for a value absent from the context, a fixed threshold under the entailment scorer catches the edit in 5 to 14% of the most verbose generator's answers and 57 to 71% of the tersest's. How many assertions an answer contains accounts for most of that gap (Section 5). A generative fact-checker narrows the gap to about 15 points without closing it (Appendix D). On human-annotated hallucinations from six other generators, every scorer catches a hallucination less often the smaller its share of the answer (Appendix E).
- Making one model more verbose by prompt alone cuts that detection from 35.1% to 10.4%, and detection falls under all three evaluators and under the fact-checker. The same prompt line also lowers the refusal rate and raises the score an unrelated document receives, so it does not vary answer length alone (Section 5.3).
- The evaluators differ in resolving power, and that is enough to move a verdict. Whether comparing the four systems returns a difference or a null depends on which evaluator scored it, in three of eight dataset–generator cells after correcting for multiple tests, and in four of eight with the fact-checker as a fourth evaluator, although no evaluator is shown to rank the systems differently (Section 7, Appendix N).
- Averaging in refusals, or choosing a different equivalence margin, changes the conclusion too (Sections 6 and 8).

2 Related Work

Retrieval quality and downstream outcome. Salemi and Zamani (2024) find that relevance labels from human annotators or from an LLM correlate only weakly with downstream RAG performance, and propose eRAG, which labels each retrieved document by what the system's own generator produces from it alone. Their strongest baseline, answer-containment labelling, is the scheme our qrels use, and it keeps a moderate Kendall's τ of 0.31 to 0.40 on the question-answering datasets. Their correlations fall as k grows and are reported at $k = 50$, against our 5, and their downstream measure is correctness from a fine-tuned 60M-parameter reader that cannot abstain. Ranking metrics may also miss how well the context covers what a long-form answer needs (Ju et al., 2025), and in mathematics question answering different embedders succeed on different queries (Chen et al., 2025). We take this gap as given and ask what a faithfulness evaluator can resolve between retrieval systems, for prompted generators that can refuse.

Faithfulness scorers. Surveys of RAG evaluation place faithfulness among the metrics for the generation component, beside relevance and accuracy (Yu et al., 2025). A systematic review of twelve RAG evaluation studies links each dimension, faithfulness included, to the metrics proposed for it and to the dataset and question types those metrics require (Knollmeyer et al., 2026). Laban et al. (2022) showed that entailment models detect document-level inconsistency once the source is split into smaller premises and pairwise scores are aggregated; both entailment aggregates we test are of this kind (Section 3.1). AlignScore (Zha et al., 2023) trains one alignment function across factual-consistency tasks, MiniCheck (Tang et al., 2024) trains small fact-checkers for grounding documents, and RAG frameworks add other dimensions or delegate the judgement to a language model (Es et al., 2024; Saad-Falcon et al., 2024; Tamber et al., 2025; Murugaraj et al., 2026), which can itself be trained for the task (Li et al., 2025). Claim-level scoring verifies atomic claims one at a time (Min et al., 2023), so its result depends on the decomposition method (Wanner et al., 2024), and decomposition trades accuracy gains against the noise it introduces (Hu et al., 2025a). ReDeEP (Sun et al., 2025) traces hallucination to attention heads and feed-forward components. Faithfulness has also been separated from correctness on the generator side, for citations

(Wallat et al., 2025) and for intermediate reasoning (Xu et al., 2025). We propose no new scorer. We compare what three common ones detect on identical inputs, and Appendix AB considers scorers we did not run.

Meta-evaluation of scorers. A separate line of work scores the detectors against human labels: TRUE (Honovich et al., 2022), AGGREFACT (Tang et al., 2023), RAGTruth (Niu et al., 2024), FaithBench (Bao et al., 2025) and the multilingual MEMERAG (Cruz Blandón et al., 2025). No single metric is best in all settings or for all error types, detection falls to about 50% on summaries where current detectors disagree, and across 37 language models hallucination metrics often fail to align with human judgments (Kulkarni et al., 2025). Chen et al. (2026) add a long-context RAG benchmark with realistic label noise. A second line constructs the errors. FactCC (Kryscinski et al., 2020) trains on rule-based edits that include entity and number swaps, GO FIGURE (Gabriel et al., 2021) measures metric sensitivity on controlled diagnostic pairs, and BUMP (Ma et al., 2023) edits reference summaries into minimal pairs. GroUSE (Muller et al., 2025) unit-tests RAG judges on generator failure modes and finds agreement with GPT-4 an incomplete proxy for passing them, and Jing et al. (2025) compare language-model and NLI scorers on synthetic unfaithful outputs. Our probe belongs to this second line, with two differences: the substituted value is absent from the retrieved context, while FactCC draws its replacements from the source document, and the edit is made to answers from the systems being compared, so each evaluator is tested on the population it would score.

Reliability of system comparisons. Xiao et al. (2023) argue for treating generation metrics as measurement instruments with stated reliability and validity. In summarization, the system-level correlation of ROUGE with human judgments is close to zero when systems differ by the small margins common in practice (Deutsch et al., 2022). Length is a known confound: response-length heuristics rival complex hallucination detectors (Janiak et al., 2025), and LLM judges favour longer responses in preference comparisons (Hu et al., 2025b). Refusals decide which answers get scored, and they depend on the system. No RAG configuration is best on both answerable and unanswerable queries across knowledge bases (Peng et al., 2025), and even frontier models fall below 50% refusal accuracy on multi-document tasks (Muhamed et al., 2026). Our results concern all three, resolving power, answer length and the evaluation population, measured for RAG faithfulness on one fixed generation grid.

3 Definitions

3.1 Faithfulness

For a generated answer a and retrieved context R = {d1, ..., dk}, the standard aggregate is:

$$F_{\max}(a, R) = \max_{d \in R} \mathrm{NLI}(d \models a) \quad (1)$$

We also evaluate a claim-level aggregate. Writing C(a) for the assertions in a:

$$F_{\mathrm{claim}}(a, R) = \min_{c \in C(a)} \max_{d \in R} \mathrm{NLI}(d \models c) \quad (2)$$

Taking the maximum over chunks makes sense, since any one retrieved document may support a claim. Taking it over claims, which (1) does implicitly, does not. We therefore score an answer by its weakest assertion, so that one unsupported claim cannot sit hidden behind several supported ones. That is a design choice and not a definition of faithfulness. A minimum is the most severe aggregation available, and it is what drives claim-min's absolute scores toward zero as answers lengthen (Section 5.3). An average over claims, or an aggregate weighted by how much content each claim carries, would encode a different and equally defensible view of what a partly supported answer is worth. We take the severe one deliberately and report what it costs. Throughout, faithfulness is measured against the generated answer and the retrieved context. The gold answer never enters it.

A natural summary would subtract faithfulness from NDCG@5 and report the gap. We do not use one, for two reasons. On Natural Questions the difference comes out negative for every model, because faithfulness (0.85) runs above retrieval quality (0.79), so the metric presumes a loss of fidelity that does not occur. And any metric of the form 1 - F/RQ reproduces the retrieval ranking whenever F is near-constant across systems, which it is here. We report absolute faithfulness instead, which has the further advantage of keeping the both-low case visible; a difference cannot tell it apart from both-high.

3.2 Correctness and retrieval hit

Correctness is measured against the gold answer, never against the context. Keeping the two apart is what lets a grounded wrong answer and an ungrounded right answer stay distinguishable. Correctness is graded by a judge model, Claude Opus 5, shown the question, the reference answers and the answer; against a blind human annotator it agrees on 99.0% of correct-versus-incorrect decisions (Appendix K). A retrieval hit means the gold answer string appears somewhere in the retrieved context. That is a strong definition of retrieval success, and Limitations sets out what it misses.

4 Experimental Setup

Two datasets are used, Natural Questions (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018), at 1,000 queries each. Four embedding models index each corpus, and retrieval takes the top five chunks of 256 tokens with 32-token overlap. Four generators answer from a byte-identical prompt at temperature 0: two proprietary (Claude Haiku 4.5, GPT-4o-mini) and two open-weight run locally (Qwen2.5-7B-Instruct, Mistral-7B-Instruct-v0.3). The two open-weight arms share a decoding path and differ only in weights, which is what makes them two generators rather than one model prompted twice. That gives 4 x 2 x 4 x 1,000 = 32,000 answers, each re-scored by every evaluator. The four systems are all-mpnet-base-v2 (contrastive), BGE-M3 (multilingual), E5-large-instruct (instruction-tuned) and text-embedding-3-small (proprietary). They reach NDCG@5 of 0.786, 0.794, 0.806 and 0.830 on Natural Questions and 0.705, 0.809, 0.824 and 0.767 on HotpotQA, a spread of 4.4 and 11.9 points. They do not rank the same way on the two datasets (Appendix A). A single prompt builder enforces parity, one user turn and no system message, since a prompt difference would confound every cross-generator comparison below.

Four further arms run off the same grid: the control of Appendix X, and three reported in Appendices Y to AA. A fifth embedding model, jina-embeddings-v3, was run over both datasets and all four generators. It serves as a control on the comparison and is not a member of the matched set. A third dataset, QASPER (Dasigi et al., 2021), was run at 819 questions over all five embedding models and three of the four generators, Mistral-7B excepted. The floor, ceiling and context-arrangement conditions and the re-ranking analysis were run on the Claude arm; the geometric analysis uses no generator. Their scope is stated where they appear and again in the Limitations.

Three evaluators are compared, on identical inputs. NLI-max is (1) with a DeBERTa-v3-large (He et al., 2023) inference model, a common choice in the work cited in Section 2. It uses the entailment probability alone. Steen et al. (2023) show that an NLI scorer predicts faithfulness better when it also uses the contradiction probability, and better again with task-adapted training and Monte-Carlo dropout. We score the out-of-the-box configuration, since the question here is what published comparisons rest on, and we test the first of those changes directly. Re-scoring every falsification case as entailment minus contradiction (Appendix C) leaves detection at the floor-anchored gate within two points of NLI-max in all eight dataset-generator cells. Claude stays the least detected generator and GPT-4o-mini the most, and AlignScore still has the larger mean drop in every cell. The contradiction term moves the scale, not what the scorer responds to. We did not try task-adapted training or Monte-Carlo dropout. A fourth, generative checker, Bespoke-MiniCheck-7B (Tang et al., 2024), was run on the falsification probe (Appendix D) and over every answer of the grid, as a fourth evaluator of the system comparison (Appendix N). Claim-min is (2) with the same model and sentence-level claim segmentation, chosen so no second generator enters the measurement loop. AlignScore (Zha et al., 2023) is an independently trained alignment model.

The answers are byte-identical across evaluators. The premises are not, and that difference is part of what the comparison measures, because premise construction follows each evaluator's intended interface. The entailment scorer sees each of the five chunks separately, plus their concatenation truncated to the scorer's own 512-token window, and takes the maximum. The claim-level scorer applies the same construction per extracted claim. AlignScore takes the concatenated context as one premise and splits it internally. Some of the difference in Section 5 is therefore premise granularity rather than evaluator sensitivity; the Limitations section bounds that share by re-scoring the same cases on the concatenated premise alone. Given AlignScore's own premise, the entailment model still responds less than AlignScore in all eight dataset-generator cells, by mean drop and

by detection. The hypothesis is normalised identically for all three, with markdown emphasis, headings and list markers stripped before scoring. That matters because Claude emits markdown in 76.0% of its attempted answers against 1.3%, 0.3% and 0.2% for Qwen, GPT-4o-mini and Mistral-7B, and the entailment model responds to it (Appendix V).

One refusal rule, from a single shared implementation, produced every evaluator score, falsification case and comparison reported below, so those sections describe the same population. The one exception is the correctness analyses of Appendix U and Appendix J: where a judge grades correctness it also classifies refusals, and those analyses take its classification. Appendix T measures how far the two classifications differ, and Appendix O reports the headline count under this rule, its bare form and the judge. An answer counts as a refusal when, after SQuAD-style normalisation, it begins with one of a fixed list of declining phrases. The rule excludes 8,055 of the 32,000 answers (25.2%), from 11.8% on NQ/Qwen to 39.0% on HotpotQA/Mistral-7B. The list is wider than the refusal sentence the prompt offers, because generators decline in their own words: Mistral-7B opens 378 refusals with "The provided context does not". A leading "Based on the provided context," is stripped before the anchored test, since Claude prepends it to 97.5% of its attempts and to some of its refusals. Without the wider list and the stripping, 759 refusals would count as attempts: 107 of Claude's, 50 of Qwen's, 602 of Mistral-7B's and none of GPT-4o-mini's.

4.1 Terms used throughout

Four words carry specific meanings below and are easy to read past.

A **cell** is one combination of the factors being held fixed, and the factors differ by analysis, so we always name the unit. A *dataset-generator cell* is one of eight, a *dataset-generator-evaluator cell* one of twenty-four, and a *dataset-generator-system cell* one of thirty-two. Unqualified, "cell" means the first.

An **arm** is one generator's worth of answers over the whole grid. The **grid** is the fixed set of 32,000 answers; the four embedding systems in it are the **testbed**.

An answer is an **attempt** if the refusal rule does not match it, and a **refusal** otherwise. Results reported as *answered-only* exclude refusals. **Paired** means the comparison is restricted to queries every system answered, so the systems are compared on the same questions.

The **spread** in a cell is the difference in mean faithfulness between the best and worst of the four systems. An evaluator's **resolving power** is how small a difference it can detect at a given sample size. The same spread may reach significance under one evaluator and not another purely because one evaluator's per-query scores are less noisy. That is a separate thing from how strongly it reacts to an error. The **floor** is what an evaluator scores for an untouched answer against an unrelated document, which is above zero and differs by evaluator, so a **floor-anchored** gate is one placed relative to it rather than at a fixed constant.

5 How Sensitive Are These Evaluators to a Known Error?

5.1 A controlled falsification

A faithfulness score is only useful if it responds when an answer stops being faithful, so we construct that condition directly. For each answer containing a numeric value that appears in the retrieved context, we replace the value with one that does not appear there, preserve the surface format, and re-score against the same chunks. Two eligibility rules carry the design: the original value must be present, or the answer was never grounded, and the replacement must be absent, or the altered answer is accidentally supported and a low score is correct behaviour.

Cases are taken deterministically, in dataset order rather than by random sampling: the first 200 eligible answers per system, dataset and generator. The scored set is therefore a prefix of an already filtered population: 6,179 cases against 9,286 eligible. HotpotQA/Mistral-7B is a full census, 595 of 595, because it has fewer than 200 eligible answers on every system. The rest run from a near-census on GPT-4o-mini (800 of 882 on NQ) to two fifths on Claude (800 of 1,946). A seeded random 200 per cell gives the same detection rates to within 2.4 points (Appendix P).

Two controls accompany it: an entity substitution, which makes a comparable edit to a named entity instead of a value, and the untouched answer scored against another query's context, which gives the floor of each evaluator's scale. Cases are rebuilt deterministically, so all three evaluators see identical inputs.

On Claude's Natural Questions answers, NLI-

		Claude Haiku 4.5			Qwen2.5-7B			Mistral-7B			GPT-4o-mini		
Dataset	**Evaluator**	Orig	Fals.	Det.	Orig	Fals.	Det.	Orig	Fals.	Det.	Orig	Fals.	Det.
NQ	NLI-max	0.906	0.852	**5.1%**	0.836	0.574	35.1%	0.818	0.567	34.3%	0.887	0.436	**56.6%**
	Claim-min	0.679	0.514	**25.1%**	0.693	0.397	48.1%	0.695	0.411	48.2%	0.863	0.390	**61.0%**
	AlignScore	0.866	0.619	**24.1%**	0.863	0.428	51.8%	0.856	0.415	51.8%	0.912	0.272	**75.4%**
HotpotQA	NLI-max	0.702	0.618	**13.9%**	0.547	0.277	54.8%	0.604	0.317	52.5%	0.577	0.186	**70.5%**
	Claim-min	0.370	0.255	**34.3%**	0.424	0.160	66.8%	0.470	0.209	59.6%	0.571	0.182	**70.6%**
	AlignScore	0.786	0.454	**48.9%**	0.749	0.255	74.5%	0.768	0.287	71.5%	0.780	0.119	**89.4%**

Table 1: Response of three evaluators to a falsified value, on identical inputs, pooled over the four matched embedding systems. Generators are ordered by how many assertions they write per answer, most first, so the two extremes are the outer blocks. "Det." is the share of answers passing a fixed 0.5 threshold before falsification and failing it after. Bold marks the detection columns of those two extremes: the same falsification, scored by the same evaluator on the same retrieved context, is caught between 5.1% and 48.9% of the time on the most verbose generator and between 56.6% and 89.4% on the tersest. n = 800 per cell, except 784 on HotpotQA/GPT-4o-mini and 595 on HotpotQA/Mistral-7B, which has the highest refusal rate in the grid and so offers the fewest answers carrying a grounded value to falsify. Scale floors, the floor-anchored gate and Wilson 95% intervals on every rate are in Appendix B.

max leaves the falsified answer at 0.852 against a floor of 0.474. That clears the fixed 0.5 gate and does not clear the floor-anchored gate for that cell, which sits at the 95th percentile of the floor distribution, 0.985. The same edit on GPT-4o-mini's answers lands at 0.436, essentially at that generator's floor of 0.506. The two open-weight arms fall between: Qwen at 0.574 against a floor of 0.470, and Mistral-7B at 0.567 against 0.492. Evaluator, context and perturbation code are identical. Only the generator that wrote the answer differs.

One case shows the comparison at its plainest. On the same Natural Questions query, with the same chunks retrieved by all-mpnet-base-v2, Claude answers "According to the provided context, season 2 of Jessica Jones is scheduled to be released on March 8, 2018." and GPT-4o-mini answers "Season 2 of Jessica Jones is scheduled to be released on March 8, 2018." The probe changes March 8 to March 13 in both. NLI-max moves Claude's answer from 0.985 to 0.983 and GPT-4o-mini's from 0.905 to 0.068. AlignScore moves them from 0.971 to 0.054 and from 0.986 to 0.018. Both answers carry a single assertion, so claim-min equals NLI-max on each. They differ only in the opening clause, which Section 5.2 returns to.

We predicted Qwen's intermediate position before that arm was run, from its assertion count alone (Section 5.2), and it holds in both datasets under all three evaluators. Mistral-7B, added later at a similar assertion count, lands in the same band: on Natural Questions the two agree to within a point under every evaluator, and on HotpotQA they differ by 2.3 to 7.2 detection points. The evaluators order consistently too, but the ordering has to name its quantity. AlignScore has the largest mean drop in all eight dataset-generator cells of the probe, and claim-min detects more falsifications than NLI-max in all eight. The full ordering — AlignScore, then claim-min, then NLI-max — holds in seven of the eight by detection rate and five of the eight by mean drop. Claim-min edges AlignScore on Natural Questions under Claude (25.1% against 24.1%), and NLI-max edges claim-min by mean drop on HotpotQA under three of the four generators. Generators differ in how far apart the evaluators are, not in which of them responds least.

What stability there is belongs to this grid rather than to the evaluators, and we did not probe outside it. The falsification probe ran on these two datasets only, so we cannot say whether the detection ordering survives a regime where retrieval fails. A different ordering does not survive it: on QASPER the evaluators' ranking by mean score inverts for one of the three generators run there (Appendix AA). An ordering that looks like a property of the evaluators can be a property of the regime.

The evaluators do not share a floor, so detection rates only become comparable once the scale is normalised, and we report two gates. On Claude's NQ cell an untouched answer scored against an unrelated context averages 0.474 under NLI-max and 0.245 under claim-min. The distribution is heavy-tailed: its 95th percentile is 0.985 under NLI-max, so an unrelated document clears a fixed 0.5 gate nearly half the time. Our second gate

sits relative to each evaluator's own floor rather than at a shared constant, which is why we call it floor-anchored, and we place it at that percentile. There, "detected" means the falsified answer is no longer better supported than an unrelated document — the same statement on every scale. Read that way NLI-max detects 50.2% of falsifications rather than 5.1%, and its gap to GPT-4o-mini narrows from roughly elevenfold to under twofold. At that gate the extremes mostly hold: Claude detects least and GPT-4o-mini most in five of the six dataset-evaluator rows. The exception is AlignScore on Natural Questions, where the floor-anchored gate compresses the four generators into 75.5% to 82.1% and Claude edges Qwen. The evaluator ordering does not carry over to that gate. There AlignScore detects least of the three on NQ/GPT-4o-mini, and claim-min falls below NLI-max on HotpotQA/Claude (Appendix B). The floor-anchored gate is a comparison device, not an operating point: it sits above the mean untouched answer.

Counterfactual substitution is also used to build training data. Ju et al. (2026) construct 99,094 examples by replacing answer-bearing entities in SQuAD and TriviaQA with type-consistent alternatives, so that following the context can be distinguished from agreeing with it by coincidence. The construction is close to ours and the purpose is not: they manufacture conflicts to train a generator toward the context, and we manufacture them to ask what a scorer does when the conflict is already there. Their scale also shows what ours is not, since our probe is 200 cases per system, dataset and generator, chosen for having a gold-standard error rather than for covering the space of errors.

5.2 Answers with more assertions are scored less sensitively

Segmenting each answer into sentence-level assertions and grouping by count gives a declining relationship in all four generators. It is not strictly monotone, and the exceptions all sit in thin tail buckets (Appendix F). Under NLI-max, Claude's response stops falling after four assertions, Qwen and Mistral-7B each rise once in the five-or-more bucket, and GPT-4o-mini rises at four assertions and falls again, on buckets of 10 and 7 answers.

On Natural Questions the four generators average 3.08, 1.77, 1.72 and 1.25 assertions per attempted answer, counted over all their attempted answers and not only the falsified ones, with median lengths of 55, 25, 20 and 15 words. That accounts for most of the cross-generator difference in Table 1. A falsified value is a smaller and smaller edit to the hypothesis as correct surrounding text accumulates, and the remainder dominates the entailment judgement. It does not account for all of it: within the single-assertion bucket the generators still differ, +0.102 against +0.385, +0.409 and +0.485. As a check on the claim-level implementation, every single-assertion answer scores identically under both aggregates (n = 3,249, largest absolute difference 2.3e-05, which is batch-padding noise). That check is what surfaced the normalisation defect in Appendix V, where it failed on all 131 formatted answers by as much as 0.729.

Claim-level aggregation is a partial remedy. As a fraction of the distance to the random-context floor, the median response on Claude's answers moves from 0.01 to 0.03, and the minimum moves the right way in 55% of cases; when the falsified assertion is not already the weakest, the minimum does not move at all. A second dilution survives inside assertions, since the four generators still differ at matched assertion count and their single-assertion answers average 23.9, 19.5, 17.2 and 16.1 words.

The open-weight generators make that residual legible. Qwen sits between Claude and GPT-4o-mini in aggregate and does not once assertion count is held fixed: at one assertion it is +0.409, against +0.485 for GPT-4o-mini and +0.102 for Claude. Its intermediate aggregate comes from its assertion *distribution*, not from intermediate per-assertion sensitivity. Mistral-7B replicates that. It averages 1.72 assertions against Qwen's 1.77 and responds by +0.385 at a single assertion, an interval that overlaps Qwen's, so two separately trained models agree both on how much they write and on what a falsified value costs them. Claude remains the outlier at +0.102. So the remaining gap is not a smooth function of verbosity. Two generators cannot tell those apart; the third makes it visible and the fourth replicates it.

Part of what sets Claude apart is how it opens an answer. It begins 97.5% of its attempted answers with an attribution clause such as "Based on the provided context," or "According to the provided context,". Qwen opens 12.4% of its attempts that way, Mistral-7B 17.6% and GPT-4o-mini 0.2%, pooled over both datasets and all four systems. Splitting the single-assertion Nat-

ural Questions answers on that clause, Qwen responds by +0.457 [+0.418, +0.495] without it and +0.129 [+0.076, +0.187] with it, and Mistral-7B by +0.426 [+0.387, +0.467] and +0.162 [+0.091, +0.238] (bootstrap 95% intervals). Claude's 91 single-assertion answers all carry it, and its +0.102 [+0.064, +0.146] sits inside both of those preamble intervals. That split is observational, so we tested it directly. On the probe's own cases, rebuilt exactly (the as-written scores reproduce the published ones to the last digit), we removed the clause from both the original and the falsified answer and scored them again. On single-assertion answers the response rises. Claude's goes from +0.102 to +0.272 on Natural Questions (paired difference +0.169, 95% interval [+0.098, +0.237]) and from +0.149 to +0.470 on HotpotQA, and NLI-max detection rises from 10.1% to 42.5% and from 20.4% to 80.6%. Mistral-7B's single-assertion answers move the same way on Natural Questions, from +0.162 to +0.291. On HotpotQA that closes the gap to the other generators' answers without the clause (+0.401 to +0.426). On Natural Questions it closes a little under half of it, and the rest is not explained. The clause also lifts the score of the untouched answer: Claude's single-assertion originals score 0.950 with it and 0.788 without it on Natural Questions. So it raises the score of a grounded answer and shields a falsified one at once. At two or more assertions removing it changes little (+0.038 for Claude on Natural Questions, and nothing distinguishable from zero in any other cell), so there the dilution described above carries the effect. The other two evaluators confirm the effect, so it is not a quirk of the entailment model. Claim-min equals NLI-max on single-assertion answers by construction. At two or more assertions removing the clause raises Claude's claim-min detection from 28.0% to 39.1% on Natural Questions and from 42.5% to 62.4% on HotpotQA, though its mean response moves by only +0.032 and +0.041. AlignScore is shielded less but measurably. Claude's single-assertion response under it rises from +0.497 to +0.656 on Natural Questions (paired difference [+0.120, +0.196]) and from +0.678 to +0.782 on HotpotQA, and by about +0.025 at two or more assertions. The clause is not stripped for any evaluator, so the treatment is identical across them. It is the same kind of choice as the markdown normalisation of Appendix V, and we report it rather than re-run the grid without it.

5.3 Verbosity, varied within one model

The gradient in Section 5.2 runs across four generators, and those generators differ in more than answer length. Two are proprietary chat models from different labs and two are open-weight models from different labs; assertion count is confounded with everything else that distinguishes them. Each further model adds a further confounded point. Varying the prompt on one model does not.

We ran Qwen2.5-7B a second time over the same queries, the same retrieved contexts and the same weights, adding one line to the prompt asking for a fuller answer. The grounding instruction and the refusal sentence are byte-identical to the standard template, so the refusal rule still matches and faithfulness is still measured against the same constraint. The arm carries its own label throughout.

The manipulation is large. Assertions per attempted answer go from 1.77 to 4.90 on Natural Questions and 1.54 to 4.99 on HotpotQA; attempted answers go from 34 and 26 words to 108 and 110. That overshoots the most verbose generator in the main grid, which averages 3.08, so the arm extends the axis rather than interpolating on it.

Sensitivity falls in all six cells (Table 2), and no confidence interval overlaps its own baseline. Under the entailment aggregate detection goes from 35.1% to 10.4% on Natural Questions and 54.8% to 17.4% on HotpotQA. The effect is not specific to the evaluator: AlignScore falls furthest in proportional terms, from 74.5% to 18.5% on HotpotQA, and its mean response to a falsified value drops from 0.494 to 0.127. The instrument Section 5.1 measures as most responsive is not exempt; it had been reading shorter answers. Nor is a generative fact-checker, which detects far more than any of the three: scoring whole answers, its detection falls from 95.9% to 65.2% and from 91.0% to 51.0% (Appendix D).

Claim-min degrades least, retaining 28.9% and 43.6% where the others fall to between 10% and 19%. That is the behaviour its construction predicts and it is here demonstrated by manipulation rather than inferred across models. Its absolute scores, though, collapse: mean 0.693 to 0.234 on Natural Questions and 0.424 to 0.076 on HotpotQA. A minimum over five assertions sits near zero whatever the answer says, so claim-min keeps its sensitivity at this verbosity while ceasing

to be usable against a fixed threshold. Resisting the structural bias and remaining thresholdable are separate properties.

What this does not show. We predicted, before running it, that an arm at 4.9 assertions would fall below the 5.1% detection of the main grid's most verbose generator at 3.08. It does not: 10.4% against 5.1%, on more assertions. Within one model, then, changing answer structure moves detection, and on this evidence it accounts for most of the distance between the two generators but not all of it. What remains is not explained by verbosity, and Section 5.2 traces part of it to a leading attribution clause. The manipulation is not perfectly clean, since the same prompt line also moves refusals and the scale floor (below). Section 5.2 reaches the same conclusion from the opposite direction, by conditioning on assertion count and finding one generator still separated from the other three. On HotpotQA the residual is small enough to disappear: 0.088 here against 0.085 there.

The verbose prompt also raised the share of the probe's cases opening with that clause, from 24.2% to 47.2% on Natural Questions and from 8.2% to 27.6% on HotpotQA, so the clause could have carried the effect. It does not. Among answers without it, NLI-max detection falls from 41.0% to 10.4% and from 57.8% to 17.1%.

Two effects that are not the one we manipulated. Refusals fall, from 11.8% to 9.8% and 29.6% to 23.8%, so the answered population is not identical between arms; the probe runs on attempted answers, but the two arms attempt slightly different questions. And the scale floor rises steeply. An untouched answer scored against an unrelated document averages 0.470 under the entailment aggregate on Natural Questions for the standard arm and 0.581 for the verbose one; on HotpotQA it goes from 0.245 to **0.509**. A five-assertion answer gives an unrelated context more chances to support something, and the aggregate takes the maximum. On HotpotQA that puts the unrelated-document baseline above the fixed 0.5 gate entirely, which is the argument of Section 5.1 for a floor-anchored gate, arrived at here by manipulation: verbosity degrades a fixed threshold twice over, by moving the falsified answer up and the floor up with it.

Evaluator	Arm	Δ	Det.	95% CI
Natural Questions				
NLI-max	standard	0.261	35.1%	[31.7, 38.8]
	verbose	0.072	**10.4%**	[8.2, 13.0]
Claim-min	standard	0.297	48.1%	[44.0, 52.2]
	verbose	0.054	**28.9%**	[22.4, 36.4]
AlignScore	standard	0.436	51.8%	[48.2, 55.4]
	verbose	0.149	**16.0%**	[13.5, 18.8]
HotpotQA				
NLI-max	standard	0.270	54.8%	[50.1, 59.5]
	verbose	0.088	**17.4%**	[14.3, 20.9]
Claim-min	standard	0.263	66.8%	[61.6, 71.6]
	verbose	0.025	**43.6%**	[31.4, 56.7]
AlignScore	standard	0.494	74.5%	[71.1, 77.7]
	verbose	0.127	**18.5%**	[15.8, 21.4]

Table 2: One model, one prompt line. Response to a falsified numeric value under the standard prompt (1.77 and 1.54 assertions per answer) and a verbose one (4.90 and 4.99), on the same queries, weights and retrieved contexts, pooled over the four matched embedding systems, $n = 800$ per row. Detection is at the fixed 0.5 gate over cases scoring above it before falsification. Every cell falls and no interval overlaps its baseline.

6 The Evaluation Population Can Manufacture a Relationship

All four generators refuse a substantial share of queries, and they refuse more often when retrieval is worse. On HotpotQA the weakest retriever draws 0.378 to 0.447 refusals and the strongest 0.224 to 0.337. A refusal is not an attempt at a grounded answer, so averaging it in carries a retrieval effect into a faithfulness number without looking as though it does.

The amount carried can be large. Pooled spreads run 0.053 to 0.085 for Claude, Qwen and GPT-4o-mini, and excluding refusals leaves 0.003 to 0.029. Mistral-7B goes the other way, from 0.039 pooled to 0.060 answered. The direction has nothing to do with refusing. It depends on how the evaluator scores the sentence each generator refuses in. On HotpotQA, GPT-4o-mini refuses with the prompt's template sentence alone, "I cannot answer based on the provided context.", in all of its refusals and Qwen in 92.5% of its own, and NLI-max scores that bare sentence at 0.93 against the retrieved chunks. Claude and Mistral-7B never stop there. They go on to say what the context lacks ("The context contains information about Ed Wood … but it does not contain …"), and those refusals average 0.31 and 0.36. Claude's and Mistral-7B's refusals score far below their attempts, so pooling drags the weakest

HotpotQA	refusal rate	pooled		answered	mean NLI-max	
	worst-best	spread	ρ	spread	refusals	attempts
Claude	0.393-0.224	0.085	+1.00	0.018	0.27-0.35	0.69-0.71
Qwen	0.378-0.241	0.053	-0.80	0.003	0.85-0.90	0.57
GPT-4o-mini	0.398-0.232	0.072	-0.80	0.029	0.90-0.94	0.62-0.65
Mistral-7B	0.447-0.337	0.039	+0.60	0.060	0.35-0.36	0.57-0.63

Table 3: Pooling refusals into the faithfulness average gives a column that tracks retrieval quality in every generator (ρ = Spearman against NDCG@5 over the four systems), but in opposite directions. With four systems ρ can only take multiples of 0.2, which is why these are round numbers. The sign is set by whether that generator's refusals score above or below its attempts: Claude's and Mistral-7B's score below (last two columns) and their pooled ρ is positive, Qwen's and GPT-4o-mini's score above and theirs is negative. This predicts the sign in all four. The *answered* column is each system's spread over its *own* answered rows, which is unpaired and therefore not the paired spread of Table 4; the two populations differ and both are reported rather than conflated. Excluding refusals shrinks the spread for three generators and enlarges it for Mistral-7B, so pooling is not a conservative simplification in either direction.

retriever down and the pooled column rises with NDCG@5. Qwen's and GPT-4o-mini's score far above theirs, so pooling lifts the weakest retriever and the same column falls. One experiment supports both readings and neither is about faithfulness. The pooled column is close to a monotone function of the refusal rate with a sign set by boilerplate, so anyone defending it cannot fall back on a consistent direction of bias. All faithfulness results here exclude refusals and are paired on the queries every system answered.

7 The Verdict Depends on the Evaluator's Resolving Power

We compare the four retrieval systems on faithfulness, refusals excluded, paired on a common query subset, with a paired bootstrap of 10,000 resamples.

The result of this section is narrow, and we state it before the table. Which evaluator is used decides whether a comparison of these systems returns a difference or a null. It does not follow that the evaluators rank the systems differently, and a direct test finds no evidence that they do: a mixed model of system by evaluator shows no detectable interaction in any of the eight cells, p from 0.31 to 0.96 (Appendix M). What separates them is resolving power, and it has two parts. AlignScore's paired per-query differences are less noisy, so it has the smallest minimum detectable effect of the three in every cell (MDE in Table 4). After that variance is divided out, its standardised effect is also larger than NLI-max's in five of eight cells (Std. in Table 4), but not on Mistral-7B, where the entailment aggregate has the larger response in both datasets. Both parts lead to the same consequence for a published comparison. The pattern holds beyond the entailment model. Scoring every answer with the generative fact-checker as a fourth evaluator (Appendix N), seven of thirty-two comparisons survive correction, in four of the eight cells. The four in Table 4 all survive the larger family, the fact-checker adds three on HotpotQA, and every Natural Questions cell stays null.

The statistic is the spread, the difference in mean faithfulness between the best and worst of the four systems, with a p-value from a paired bootstrap over that pair re-centred at zero (Dror et al., 2018). We report it as a deliberately extreme sensitivity check, and not as the loss a practitioner should expect. Someone who adopts one of these systems stands to lose the distance from it to the best of the four. That equals the spread only when the system they adopted is the worst one, and is smaller under the other three choices. The spread is the largest shortfall the choice can produce, which is the right quantity for asking whether the choice can matter at all and the wrong one for asking what it costs on average. It also summarises two systems picked for being the extremes, so its p-value is not adjusted for that selection and the two intermediate systems never enter it. Appendix M runs two tests that do pay for the selection. The permutation test retains two of the four comparisons, in two dataset-generator cells, and the rank-based omnibus retains one. The direction across all four systems is in Appendix W.

Under the common evaluator one of the eight cells shows a detectable difference, on Mistral-7B; under the evaluator Section 5 measures as most responsive to known errors, two do. All four surviving comparisons are on HotpotQA, which has the wider retrieval spread, and all four Natural Questions cells are null. That may be a property of how the Natural Questions sample is drawn rather than of the evaluators. It retains only queries whose annotated answer falls inside the retained context window, which leaves a constrained single-hop task with less for an embedding model to get wrong. We cannot separate that from the two datasets differing in hop structure.

		Natural Questions				HotpotQA			
Generator	**Evaluator**	Spread	MDE	Std.	p (Holm)	Spread	MDE	Std.	p (Holm)
Claude Haiku 4.5	NLI-max	0.0107	0.021	0.051	1.000	0.0232	0.046	0.062	1.000
	Claim-min	0.0216	0.038	0.057	1.000	0.0345	0.050	0.085	0.898
	AlignScore	0.0114	0.012	0.094	0.154	0.0164	0.025	0.082	1.000
Qwen2.5-7B	NLI-max	0.0055	0.025	0.022	1.000	0.0123	0.060	0.026	1.000
	Claim-min	0.0076	0.032	0.024	1.000	0.0181	0.049	0.046	1.000
	AlignScore	0.0072	0.016	0.045	1.000	**0.0406**	0.035	0.145	**0.024**
GPT-4o-mini	NLI-max	0.0130	0.022	0.066	1.000	0.0267	0.041	0.081	1.000
	Claim-min	0.0151	0.024	0.071	1.000	0.0378	0.040	0.116	0.140
	AlignScore	0.0080	0.015	0.062	1.000	**0.0461**	0.037	0.155	**0.012**†
Mistral-7B	NLI-max	0.0167	0.032	0.056	1.000	**0.0796**	0.061	0.177	**0.007**†‡
	Claim-min	0.0155	0.039	0.042	1.000	**0.0699**	0.064	0.149	**0.038**
	AlignScore	0.0030	0.018	0.017	1.000	0.0305	0.037	0.114	0.333

Table 4: The same comparison under three evaluators, Holm-corrected over the twenty-four comparisons in the table. Spread is the difference in mean faithfulness between the best and worst of the four systems. MDE is the minimum detectable effect at 80% power for a two-sided paired test at $\alpha = 0.05$, and Std. is the spread divided by the standard deviation of the same pair's per-query differences, which removes each evaluator's variance (Appendix L). Spread and MDE are on the evaluator's own scale; Std. is unitless and is the only one of the three comparable across evaluators. Bold marks the four comparisons that change verdict at alpha = 0.05 after correction, in three of the eight dataset-generator cells, all on HotpotQA. †: also survives a permutation test that re-selects the extreme pair in every resample; ‡: also survives a Friedman test over all four systems (Appendix M). Queries, answers, refusal handling and test are identical throughout. Paired n is 784, 790, 628 and 692 on NQ and 517, 499, 510 and 423 on HotpotQA. With 10,000 resamples the smallest attainable non-zero p is 1e-4, so the cell at p = 0.0003 uncorrected is 3 extreme resamples of 10,000, not a clamped floor.

The MDE column of Table 4 reports what each cell could have detected: the minimum detectable effect at 80% power is 0.012 to 0.064, above the observed spread in twenty of the twenty-four dataset-generator-evaluator cells, and it is lowest under AlignScore in all eight. Part of the difference between evaluators here is resolving power at fixed n, not only response to a known error. The Std. column separates the two: dividing each spread by its own paired standard deviation removes the variance term, and AlignScore's standardised effect still exceeds the entailment aggregate's in five of eight cells. So lower variance is not the whole of it: in five cells AlignScore also responds more strongly relative to its noise. On Mistral-7B the ordering reverses and the entailment aggregate is the more responsive of the two in both datasets, consistent with that arm breaking on HotpotQA under NLI-max and claim-min and not under AlignScore. A result reported only under the first would describe the resolution of the instrument, not a property of the systems. None of the four is on Claude, whose answers the entailment aggregate resolves least well.

Each added generator made the correction stricter — twelve comparisons, then eighteen, now twenty-four — and the count rose each time. The design does not guarantee that. Adding generators also qualified what had been the sharpest result here: under AlignScore on HotpotQA, three of the four generators rank the four systems identically (all-mpnet < text-embedding-3-small < BGE-M3 < E5-instruct), where under NLI-max no two of those three agree. Mistral-7B is the exception, ranking them text-embedding-3-small < all-mpnet < E5-instruct < BGE-M3, and it is also the one generator whose AlignScore spread is not distinguishable from zero (Table 4), so that ordering is read off differences inside its own noise. Read the agreement carefully in any case. The generators are independent of each other but answer from the *same* retrieved contexts, so this is three generators agreeing about a property of the shared retrieval, not three replications. What it establishes is that one evaluator resolves that property consistently across most generators and another does not, on identical inputs.

The count of four is not invariant to the correction either. Three comparisons sit either side of alpha = 0.05 depending on which are treated as one family. NQ/Claude under AlignScore is p = 0.0081 uncorrected, adjusted to 0.154 over the twenty-four in the table, 0.049 over the eight sharing an evaluator, 0.024 over the three in its

own cell. HotpotQA/GPT-4o-mini under claim-min is p = 0.0070, adjusted to 0.140, 0.049 and 0.014. HotpotQA/Mistral-7B under AlignScore is p = 0.0185, adjusted to 0.333, 0.093 and 0.019. Correcting within evaluator makes the count six and correcting within cell makes it seven. Correcting within evaluator is defensible, since the eight NLI tests establish which cells are null and so precede the question rather than competing with it. We still report the twenty-four-comparison family, because it is the most conservative and was fixed before the correction was applied, and choosing a smaller one after seeing that it restores cells is a selection we could not account for. We give the alternative as well as the choice, because a count that moves with the multiplicity family depends on an analysis decision in the same way this paper reports for the choice of evaluator.

The statistic moves it too, and in the less comfortable direction. Appendix M replaces the paired test on the extreme pair with a permutation test that re-selects the extremes in every resample, so the cost of choosing them is paid: two comparisons survive Holm rather than four, HotpotQA/GPT-4o-mini under AlignScore and HotpotQA/Mistral-7B under NLI-max. A rank-based omnibus over the same four systems leaves one comparison, the second of those. The two counts answer different questions. A practitioner comparing their adopted model against the best alternative is exposed to the extreme pair, while the omnibus asks whether the four differ at all. The paired figure is the most permissive of the three, so we report all of them. The same appendix tests whether the evaluators disagree about the four systems, rather than merely returning different verdicts about them, and finds no detectable interaction in any of the eight cells. Read with the Std. column of Table 4, the defensible claim is that these evaluators resolve a common pattern with different resolving power. That leaves the consequence for a published comparison unchanged and makes the mechanism narrower than evaluator disagreement.

8 What an Equivalence Claim Requires

Five of the eight cells in Table 4 show no detectable difference under any evaluator. Table 4 puts the minimum detectable effect above the observed spread in twenty of the twenty-four dataset-generator-evaluator cells, so an effect the size of the one we measured would usually not have been detected. A null here is not evidence of equivalence, so we test for equivalence directly with two one-sided tests (Schuirmann, 1987). TOST turns on the margin, and the margin is a judgement about what difference would matter. We sweep it from ± 0.01 to ± 0.10 and report the curve (Appendix H).

The curve is steep. On NQ/Claude under NLI-max no pair is equivalent at ± 0.01, four of six are at ± 0.02 and all six at ± 0.03. HotpotQA needs a far wider margin. No pair there is equivalent at ± 0.01 under any evaluator, and below ± 0.05 only four of its eighteen generator-evaluator rows contain any equivalent pair at all: AlignScore on Claude, which reaches five of six at ± 0.03, and AlignScore on Mistral-7B and both NLI-based aggregates on GPT-4o-mini, with one each. At the other end, HotpotQA under Mistral-7B is the only cell where ± 0.10 still leaves a pair non-equivalent, under NLI-max and under claim-min. Stating equivalence means choosing a column, and the choice does most of the work.

Section 5 gives a way to choose it. Falsifying one grounded value moves NLI-max by 0.049 to 0.095 on Claude's answers, so ± 0.05 on that evaluator is about the size of one fabricated fact, and equivalence at that margin says the systems differ by less than one fabrication. The anchor is evaluator-specific and so is the verdict: the same falsification moves claim-min by 0.093 to 0.179 and AlignScore by 0.243 to 0.364 on those answers, and at ± 0.03 on NQ/Claude, NLI-max calls all six pairs equivalent while claim-min calls two. An equivalence claim has to name its evaluator, and say what the margin is worth on that evaluator's scale.

The anchor fixes the units, not the tolerance. Saying ± 0.05 is about one fabricated fact on this evaluator does not establish that one fabricated fact is the right bar, and nothing in our data can establish that. A system summarising product reviews and a system reading a drug label have different tolerances for the same measured difference, and the second may want a margin an order of magnitude tighter. What the anchor removes is the case where a margin is chosen because it is conventional and its magnitude on the chosen evaluator's scale is never checked. Reporting the curve leaves the choice with the reader, who knows the application.

One limit on the anchor runs in a direction we

can name. It is calibrated on errors we constructed, and Sinha (2025) report that naturally occurring hallucinations are harder to separate than constructed ones, because a real hallucination stays semantically closer to a faithful answer (Limitations). Where that holds, one real fabrication moves the score by less than the 0.049 to 0.095 we measure. A margin set at one constructed fabrication is then worth more than one real fabrication, and equivalence declared at it tolerates more than the anchor appears to allow. On RAGTruth's annotated hallucinations (Appendix E) it holds for AlignScore and the fact-checker, which separate natural errors less well than ours by 0.09 to 0.14 in AUROC. For NLI-max, the evaluator the anchor is set on, the two are about equally hard. That comparison crosses datasets and generators and has no paired drop, so it cannot size the error. Where there is a difference its sign makes a margin calibrated on constructed errors permissive, not strict. Anyone holding labelled natural errors from their own traffic should re-anchor the margin on those instead.

9 Discussion

For work reporting RAG faithfulness, our measurements support six practices. Appendix AC gives the evidence and caveats for each.

- Test a faithfulness threshold on falsified answers from the generator it will guard. A fixed gate under the common entailment scorer catches 5.1% of falsified values on the most verbose generator and 56.6% on the tersest, so a threshold validated on one generator's answers does not transfer to another's. A generative fact-checker catches far more, 80.6% to 97.5%, and still loses 31 and 40 points when the same model writes more, so a stronger checker does not remove the need for the test (Appendix D). Score untouched answers against unrelated contexts as well, since that floor moves with verbosity (Appendix AD).

- Name the evaluator. "Faithfulness did not differ" is incomplete: in three of our eight cells the same data returns a difference under one evaluator and a null under another. The evaluators are not shown to rank the systems differently. They differ in resolving power, and the instrument is what the reader needs in order to know what a null is worth.

- Report the evaluation population, and whether membership depends on the system being compared. Pooling refusals here produced a column tracking retrieval quality in all four generators, in opposite directions, and the direction is predictable from whether that generator's refusals outscore its attempts (Table 3).

- When individual answers are being checked for a fabricated value, prefer claim-level aggregation to a maximum over the whole answer, and treat it as a partial remedy. It lifts the verbose generator's detection on Natural Questions from 5.1% to 25.1%, still a minority, and its absolute scores fall toward zero as answers lengthen, so a fixed gate carried over to it would reject almost every verbose answer.

- For symbolic content, put a deterministic check in front of the evaluator. That it catches our falsifications is circular, since the probe defines the substituted value as absent from the context. What we measure is its cost, 0.8% to 5.2% of untouched answers flagged, against detection of 5.1% to 89.4% for the three evaluators of the main text and 80.6% to 97.5% for the fact-checker of Appendix D, on the same cases.

- Calibrate the margin against a measured effect before claiming equivalence. A falsification experiment of this kind costs nothing beyond re-scoring. On natural hallucinations constructed errors are the easier case for the stronger scorers and the harder case for none, so a margin anchored on them errs toward permissive (Section 8).

What this paper adds is the controlled minimal pair. One value is replaced in an otherwise byte-identical generation, and the same case goes to all three evaluators. What is measured is each evaluator's response to a known error on identical input, not its agreement with a human label on some other corpus. That covers numeric values, with entities as a second error type (Appendix R), four generators and two datasets. Appendix E checks the dilution mechanism on human-annotated natural hallucinations.

10 Conclusion

Injecting one known error into answers from a fixed 32,000-answer generation grid shows that a faithfulness score's sensitivity depends on the generator it scores. A falsified grounded value moves the common entailment aggregate by 0.054 on one generator, 0.251 and 0.261 on two more and 0.451 on a fourth, and how many assertions an answer contains accounts for most of that gap; making one model more verbose by prompt alone reproduces it. A generative fact-checker detects far more and narrows the gap between generators to about 15 points, but loses as much detection as the entailment aggregate when the same model writes more. The evaluators also differ in resolving power. Varying only the evaluator changes the conclusion about whether embedding choice affects faithfulness in four of twenty-four comparisons, which fall in three of eight dataset-generator cells, or in two comparisons in two cells under a test that pays for selecting the pair compared. A generative fact-checker scored as a fourth evaluator raises this to seven of thirty-two comparisons, in four of the eight cells. The evaluators are not shown to disagree about the systems, but a difference in resolving power is sufficient to move a published verdict. A faithfulness result that does not name its evaluator, population, aggregation and equivalence margin is not reproducible in the sense that matters: another group making different standard choices on the same generations would reach a different conclusion.

Code and Data Availability

The pipeline, the evaluator interfaces, the falsification probe, the refusal rule and every analysis script behind the tables in this paper are available at `https://github.com/Shakhriyorbek/rag-faithfulness`. The repository includes a map from each method claim in this paper to the function that implements it, and scripts that regenerate each table from stored checkpoints. This matters for a paper whose central recommendation is to re-run the probe on one's own outputs: the probe is roughly 200 cases per cell and needs no API access once generations exist. The generated answers and their scores are retained and can be shared on request; we do not redistribute the source datasets, which are publicly available.

Limitations

What we can claim is bounded by the grid we measured on, and by the graders that decide who belongs to each population.

- Four embedding models, two English datasets, four generators. Two things this bounds should be separated, because our controls move one and not the other. The *axis* the mechanism in Section 5.2 runs along is assertions per answer. As prompted, the four generators span 1.25 to 3.08 assertions per attempted answer on Natural Questions and 1.08 to 2.39 on HotpotQA. The prompt-varied arm of Section 5.3 carries one of them to 4.90 and 4.99 on the same two datasets, where the effect continues in the direction the mechanism predicts. The span of the axis we have measured is therefore about twice the generators' own span on Natural Questions and three times on HotpotQA. The *population* of models is four, and no control here changes that. Varying a prompt holds the weights fixed. That is what makes it a clean test of the confound between length and model identity, and equally what stops it saying anything about models trained differently, so the prompt-varied arm does not raise the population to five. All four are instruction-tuned chat models from broadly similar recipes, and two of them are 7B open-weight models. Part of the residual that answer structure fails to explain is an attribution clause that one generator uses on almost every answer. Removing it closes the single-assertion gap on HotpotQA and a little under half of it on Natural Questions, where the remainder is unexplained. Whether any of this holds for a much larger model, a much smaller one, or a different training regime is untested.

- Correctness is graded by the judge over the thirty-two cells of the matched grid and nowhere else. Within that grid it covers all four generators, so Appendix U and every figure conditioned on correctness are four-generator results. Outside it the judge was not run. The jina control (Appendix X), QASPER (Appendix AA) and the floor, ceiling and arrangement conditions (Appendix Y) are graded by string contain-

ment instead. The item below shows that containment is biased in both directions, by a margin that varies with the generator and the dataset. The accuracy figures reported for those arms are therefore not directly comparable with the matched grid's, and we do not pool them. Nothing in Sections 5, 7 or 8 depends on that grading.

- The grader we use in place of string overlap is biased, and we chose it knowing the bias. Exact match is 0.000 in every cell we measured, because the generators do not emit bare answer spans. String containment is the obvious fallback, and it bounds accuracy in neither direction. Measured against the judge it understates accuracy by 5 to 16 points in twenty of the twenty-four cells where both signals exist. It overstates accuracy by 2 to 6 points in the four HotpotQA cells under Claude, where a long answer mentions a reference string while asserting something else. The judge is itself a language model reading model output. Checked against a human annotator who could not see its verdicts, on 200 answers drawn evenly across the eight dataset-generator cells, it agrees on 98.5% of labels ($\kappa = 0.97$) and on 99.0% of correct-versus-incorrect decisions (Appendix K). That annotator is the author, and a second would be needed to know how far two people agree on the same answers.

- Retrieval hit is defined as the gold answer string appearing in the retrieved context. This is a strong definition, and it does not separate retrieval failure from failures of composition, entity resolution or inference once the evidence is present.

- The Natural Questions sample retains only queries whose annotated answer falls inside the retained context window. This makes the retrieval task more constrained than open-domain RAG, where evidence may be distributed, partial or implicit. It may also be why all four Natural Questions cells are null under every evaluator while all three cells with a surviving comparison are on HotpotQA. A constrained single-hop task leaves less for an embedding model to get wrong, so the systems have less to differ about. We cannot separate that from the datasets differing in hop structure, and do not claim to.

- Pairing on the queries every system answered removes exactly those queries where retrieval quality determined whether an answer was produced, so the comparison is conditional on a subset selected partly by the systems being compared. Substituting each system's own answered rows moves the spread by at most 0.022 and reorders the systems in eleven of the twenty-four combinations, three of eight under NLI-max. It leaves the ordering unchanged in three of the four comparisons that survive correction. The exception is HotpotQA/Mistral-7B under claim-min, whose ordering does change, so that comparison is in part conditional on the pairing and we do not read an ordering off it. The pairing cannot explain disagreement between evaluators in any case, because it is identical across them.

- Where we do detect a difference it is small: 0.041 to 0.080 in mean faithfulness across the four surviving comparisons, all on HotpotQA, whose systems span 11.9 NDCG@5 points. They are detectable, but they are not large.

- The three evaluators receive byte-identical answers but not identical premises. The entailment scorer is given each of the five retrieved chunks separately, plus their concatenation truncated to the model's window, and the maximum is taken. The claim-level scorer applies that construction per extracted claim. AlignScore is given the concatenated context as one premise and performs its own splitting. Each scorer is used as its authors intend, but that means some of the difference reported in Section 5 comes from premise granularity and not from evaluator sensitivity. We bound that share by re-scoring the same cases with the same entailment model on the concatenated premise alone. We ran that control in all eight dataset-generator cells, and the answer depends on which quantity is asked about, so we give both. As a fraction of the NLI-max-to-AlignScore gap in mean drop it runs from -0.55 to +0.20, negative in four cells: by that measure a concatenated premise makes the entailment model *less* responsive. As a fraction of the gap in detection at the

0.5 gate it runs from +0.01 to +0.62 and is positive in all eight, because the concatenated premise also lowers the untouched answer's score and so moves more answers across the gate. What holds under both is the comparison that matters: AlignScore exceeds the same entailment model given AlignScore's own premise in every one of the eight cells, by mean drop and by detection. Premise granularity accounts for at most about three fifths of the detection gap and none of the mean-drop gap in half the cells, so it does not account for AlignScore's advantage. The hypothesis is normalised identically for all three scorers (Appendix V). Before that normalisation was applied during development, the difference alone decided one cell.

- Which answers count as refusals is fixed by a rule rather than given by the data, and that rule sets the evaluation population for every result from Section 5 onward. It is a researcher degree of freedom as much as a finding. Swapping one defensible detector for another moves the rows nearest the boundary between refusing and attempting, and carries the uncorrected p-value across the conventional threshold in two of the twenty-four comparisons (Appendix T). Neither of those two survives correction for multiple comparisons, but the population every reported number rests on still depends on the choice. The rule used here matches a wider list of declining openings than the prompt's own refusal sentence and strips a leading preamble first; without those two extensions 759 refusals would be graded as attempts, 602 of them Mistral-7B's. Appendix O reports the headline count under each of the three defensible rules available to us. The rule used here, the same rule without its two extensions, and the judge's classification all select the same four comparisons, so the uncorrected p-values move with the rule while the corrected count does not. Against the human annotator of Appendix K the rule never calls an attempt a refusal, but it misses 7 of 61 refusals, most of them answers that give part of an answer before declining, so it leaves some refusals in the attempted population.

- The falsification cases are the first 200 eligible answers in dataset order for each system, dataset and generator, rather than a random sample of the eligible answers: 6,179 cases scored against 9,286 eligible across the grid. Eligibility requires a grounded numeric value in the answer together with a replacement value absent from the context, so the scored set is a prefix of an already filtered population. Cells whose eligible count is near 200 are close to a census, and HotpotQA/Mistral-7B is one; the largest cells are sampled at about two fifths. Appendix P tests whether that prefix is exchangeable with the eligible set in two ways. Detection shows no trend with position that survives correction over the eight tests. A seeded random 200 per cell, scored with four scorers, moves detection at 0.5 by at most 2.4 points (median 0.4), and every ordering the paper reports holds on it.

- The repository configures seven embedding models. Four carry the matched comparison — all-mpnet-base-v2 (contrastive), BGE-M3 (multilingual), E5-large-instruct (instruction-tuned) and text-embedding-3-small (proprietary) — chosen to span pretraining paradigms and to include one closed-source arm. A fifth, jina-embeddings-v3 (Sturua et al., 2024), was run over both datasets and all four generators and is reported in Appendix X as a control on the matched set, not a member of it, because it fails the matched-quality precondition on Natural Questions. GTE-large and Instructor-XL remain unrun: each additional model is a further generation pass over every dataset and generator, and the grid is a testbed for the evaluator comparison rather than a survey of embedding models.

- The floor, ceiling and context-arrangement conditions of Appendix Y were run on the Claude arm only. The geometric analysis of Appendix Z covers the four matched systems and uses no generator, and re-ranking was run on one embedding system with Claude. Neither the anchor conditions nor re-ranking has been shown to hold across generators, and the re-ranking result in particular is a single operating point ($\lambda = 0.6$) on a single system, so it bounds what that intervention achieved here. It says nothing about what such interventions can achieve in general.

- The oracle condition supplies answer-bearing chunks only. That is narrower context than top-five retrieval, so it ceilings evidence selection and not accuracy, and retrieval exceeds it on Natural Questions. Defining the ceiling as gold passages in full would raise it, at the cost of comparability with what retrieval actually produces.

- QASPER is reported as a boundary condition rather than a third dataset. Retrieval fails on it under every system tested, so the population it yields cannot test a claim about systems at matched retrieval quality, and the inverted evaluator ordering we observe there is measured on answers produced largely without supporting evidence.

- All three datasets are English, and the two that carry the main grid are built on Wikipedia. The no-retrieval condition shows 32.2% of Natural Questions is answerable from parametric memory alone, so the Wikipedia dependence is measured rather than hypothetical.

- The detection rates here bear on deployment. A system that gates answers on a fixed grounding threshold, using the common entailment aggregate on a verbose generator, would pass 94.9% of the answers in which we falsified a grounded value on Natural Questions. Anyone operating such a gate in a setting where a wrong number matters – an invoice, a dosage, a contract term – should measure its detection rate on their own generator's output before relying on it, because the rate is a property of that pairing and not of the threshold. We did not test deployment-scale traffic, and the probe injects errors rather than observing natural ones. Sinha (2025) report embedding-based detection separating constructed hallucinations at a 0% false-positive rate and naturally occurring ones at 100%, on the grounds that real hallucinations stay semantically closer to a faithful answer than constructed ones do. On RAGTruth's human-annotated hallucinations (Appendix E) we find that direction for AlignScore and the fact-checker, whose AUROC is 0.09 to 0.14 lower than on our falsified answers, no clear difference for the two entailment aggregates, and nothing near Sinha's extreme. The comparison crosses datasets, generators and error types, so it gives a direction and not a size. Where the direction holds, the detection rates here are upper bounds on what the same gate would catch in deployment, and the verbose-generator figures are already low. Sinha's result is from an unrefereed preprint. It bounds the equivalence margin in that same direction, since the margin is anchored on the same constructed errors (Section 8).

- The correct-versus-incorrect faithfulness comparison rests on 44 to 134 attempted wrong answers per cell. Excluding refusals and grading correctness with a judge both reduce that population, and the resulting confidence intervals are ± 0.064 to ± 0.129, that is 0.129 to 0.259 wide. The tests reported in Appendix U bear on the general claim and do not resolve individual systems. Wrong means wrong against the dataset's reference. In 8 of the 18 answers the human annotator of Appendix K graded incorrect, the annotator noted that the answer looked defensible, so the wrong side of the comparison includes some answers a reader might accept.

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A Retrieval quality of every system run

B Falsification response in full

C Entailment minus contradiction

Steen et al. (2023) score an NLI model by subtracting the contradiction probability from the entail-

Embedding model	Paradigm	NDCG@5 (NQ)	NDCG@5 (HotpotQA)
all-mpnet-base-v2	contrastive	0.786	0.705
BGE-M3	multilingual	0.794	0.809
E5-large-instruct	instruction-tuned	0.806	0.824
text-embedding-3-small	proprietary	0.830	0.767
spread, matched four		*4.4 pts*	*11.9 pts*
jina-embeddings-v3	distilled	0.685	0.772
spread, all five		*14.5 pts*	*11.9 pts*

Table 5: Retrieval quality of every system run. The four above the rule carry the matched comparison. Their ranking differs between datasets, and their spread is far larger on HotpotQA. jina-embeddings-v3 is reported separately (Appendix X). It sits inside the band on HotpotQA, but on Natural Questions it is the weakest system by ten points, where it triples the spread and fails TOST equivalence against all four at ± 0.02.

ment probability. We re-scored the cases of Table 6 that way, with the same model, the same six premises per answer and the same maximum. The score is rescaled from $[-1, 1]$ to $[0, 1]$, so that a 0.5 gate means entailment and contradiction are equal. The query identifiers match Table 6 in every cell, so the cases are the same.

Three things follow from Table 7. First, the contradiction term makes the entailment model somewhat more responsive in its own units: its mean drop is 1.17 to 1.29 times NLI-max's. Even on that native scale, which has twice the range, it stays below AlignScore's mean drop in all eight cells. Second, at the floor-anchored gate the two entailment scorings detect the same share of falsifications, within two points in every cell and with overlapping intervals. At the 0.5 gate E−C detects far less, 0.1% to 39.1%, but the reason is the gate rather than the scorer. The rescaled score of an untouched answer against an unrelated context averages 0.53 to 0.72, so a fixed 0.5 gate sits below the scale floor. Third, the generator pattern is unchanged. Claude is the least detected generator and GPT-4o-mini the most in all eight rows, at both gates. In the verbosity arm, detection falls from Qwen to its verbose variant on both datasets: from 5.8% to 1.0% and 24.6% to 5.0% at the 0.5 gate, and from 72.9% to 39.7% and 68.8% to 29.3% at the floor gate. A better-configured entailment score therefore changes how a fixed threshold should be placed. It does not change which answers the scorer can see an error in.

D An LLM fact-checker

The three evaluators of the main text are an entailment model under two aggregations and AlignScore, a RoBERTa classifier. To test whether the generator dependence reaches a generative checker, we re-scored every case of Table 6, and of the verbosity arm, with Bespoke-MiniCheck-7B, the 7B member of the MiniCheck family (Tang et al., 2024).[1] It reads the whole retrieved context and a claim in one prompt and answers Yes or No; the score is the probability of Yes. We use the published system and user prompts verbatim. The five retrieved chunks are far shorter than the model's document window, so the context is one document. On the example in the authors' repository it returns 0.987 and 0.009 against their 0.984 and 0.011; the remaining difference is numerical precision.

MiniCheck is designed to check one sentence at a time, and its reference implementation splits the claim into sentences and takes the minimum. We report both readings from the same pass. *Whole* scores the full answer as one claim, the single-number setting of the main text. *Per sentence* is the published aggregation, the analogue of claim-min. The query identifiers match Table 6 in every cell.

It detects far more. Scoring the whole answer, MiniCheck catches 80.6% to 97.5% of falsified values across the four generators, against 5.1% to 70.5% for NLI-max and 24.1% to 89.4% for AlignScore on the same cases. Its mean drop, 0.524 to 0.765, is larger than AlignScore's in all eight cells. It also passes 81% to 94% of untouched answers at the gate, so the detection is not bought by rejecting everything. The entity substitution gives the same picture, 73.5% to 98.8% at the 0.5 gate.

The generator dependence shrinks and does not vanish. Claude remains the least detected generator on both datasets, at 82.7% [79.7, 85.2] on Natural Questions and 80.6% [77.5, 83.4] on HotpotQA. The distance to the most detected generator is about 15 points on each dataset, where it is 52 and 57 under NLI-max.

Verbosity still costs detection. In the within-model verbosity arm, whole-answer detection falls from 95.9% to 65.2% [61.6, 68.6] on Natural Questions and from 91.0% to 51.0% [47.3, 54.7] on HotpotQA, and the mean drop from 0.626 to 0.439 and 0.587 to 0.349. That is 31 and 40 points, against 25 and 37 for NLI-max. The checker starts

[1] `https://huggingface.co/bespokelabs/Bespoke-MiniCheck-7B`, revision `1ed7786`, run locally in 16-bit precision.

Dataset / generator	Evaluator	Orig	Falsified	Floor	Detected @0.5	Detected @floor gate
NQ / Claude Haiku 4.5	NLI-max	0.906	0.852	0.474	5.1% [3.8-7.0]	50.2% [45.7-54.7]
	Claim-min	0.679	0.514	0.245	25.1% [21.7-28.9]	58.3% [53.3-63.1]
	AlignScore	0.866	0.619	0.331	24.1% [21.3-27.3]	77.9% [74.4-81.1]
NQ / Qwen2.5-7B	NLI-max	0.836	0.574	0.470	35.1% [31.7-38.8]	71.0% [66.1-75.4]
	Claim-min	0.693	0.397	0.378	48.1% [44.0-52.2]	79.0% [74.6-82.8]
	AlignScore	0.863	0.428	0.240	51.8% [48.2-55.4]	75.5% [72.1-78.7]
NQ / Mistral-7B	NLI-max	0.818	0.567	0.492	34.3% [30.8-38.0]	74.1% [69.4-78.3]
	Claim-min	0.695	0.411	0.409	48.2% [44.1-52.4]	83.5% [79.1-87.2]
	AlignScore	0.856	0.415	0.261	51.8% [48.2-55.4]	77.3% [74.0-80.3]
NQ / GPT-4o-mini	NLI-max	0.887	0.436	0.506	56.6% [52.9-60.1]	85.2% [81.9-88.1]
	Claim-min	0.863	0.390	0.490	61.0% [57.4-64.6]	91.0% [88.1-93.2]
	AlignScore	0.912	0.272	0.160	75.4% [72.2-78.4]	82.1% [79.2-84.7]
HotpotQA / Claude	NLI-max	0.702	0.618	0.427	13.9% [11.3-17.0]	49.7% [42.3-57.2]
	Claim-min	0.370	0.255	0.161	34.3% [29.0-40.0]	48.9% [41.8-56.1]
	AlignScore	0.786	0.454	0.475	48.9% [45.2-52.5]	93.7% [90.7-95.8]
HotpotQA / Qwen2.5-7B	NLI-max	0.547	0.277	0.245	54.8% [50.1-59.5]	69.6% [64.0-74.7]
	Claim-min	0.424	0.160	0.178	66.8% [61.6-71.6]	76.9% [71.3-81.7]
	AlignScore	0.749	0.255	0.304	74.5% [71.1-77.7]	96.6% [94.4-98.0]
HotpotQA / Mistral-7B	NLI-max	0.604	0.317	0.329	52.5% [47.3-57.6]	72.1% [65.2-78.1]
	Claim-min	0.470	0.209	0.257	59.6% [53.7-65.2]	78.2% [71.3-83.8]
	AlignScore	0.768	0.287	0.352	71.5% [67.4-75.3]	95.8% [92.9-97.6]
HotpotQA / GPT-4o-mini	NLI-max	0.577	0.186	0.201	70.5% [66.2-74.6]	83.8% [79.7-87.2]
	Claim-min	0.571	0.182	0.204	70.6% [66.2-74.6]	84.2% [80.1-87.6]
	AlignScore	0.780	0.119	0.234	89.4% [86.8-91.6]	97.5% [95.8-98.5]

Table 6: Table 1 with the scale floors, the floor-anchored gate and Wilson 95% intervals, in the same generator order. "Floor" is the mean score of an untouched answer against an unrelated query's context; the floor gate is the 95th percentile of that distribution, per cell and evaluator. Counted over all eight dataset-generator cells, AlignScore has the largest mean drop in every one and claim-min detects more than NLI-max in every one, while the full three-way ordering holds in seven cells by detection and five by mean drop.

from a much higher level and loses as much when the same model writes more. The untouched pass rate rises slightly, from 83% and 82% to 88% in both datasets, so this is lost sensitivity rather than a gate that verbose answers no longer clear.

Per-sentence aggregation repeats the claim-min trade. Taking the minimum over sentences keeps detection at 94% to 99% everywhere, the verbosity arm included. It pays for that in the untouched answers. Only 32% and 16% of the verbose arm's untouched answers pass the 0.5 gate, against 81% and 72% for the standard arm, and their mean score falls from 0.738 to 0.397 and from 0.668 to 0.259. Detection is high on the few answers that still pass, and a fixed gate would reject most verbose answers before any error was introduced. That is the trade Section 5.3 reports for claim-min, now on a different model.

The floor-anchored gate does not suit this scale. MiniCheck scores an untouched answer against an unrelated context near zero, with a mean of 0.016 to 0.044 on the main grid, so the floor-anchored gate sits between 0.06 and 0.16 for whole answers and below 0.06 per sentence. Passing it means scoring above all but 5% of unrelated documents, and a falsified answer counts as detected only if it falls as low as one. Detection at that gate is 41.9% to 63.7% whole and 4.0% to 47.4% per sentence. We read MiniCheck at the 0.5 gate, which is also the decision rule of its reference implementation.

E Natural hallucinations

Every detection rate in the main text comes from an error we constructed. To see whether the pattern holds for errors a generator makes on its own, we ran the scorers of Table 8 over the question answering part of RAGTruth (Niu et al., 2024). It holds answers to MS MARCO questions from six generators, each grounded on three passages, with hallucinated spans marked by annotators. None of the six is one of our generators; the Mistral-7B-Instruct model in RAGTruth is an earlier release than ours. We kept answers RAGTruth marks as of good quality and dropped refusals under our own rule (Section 4) or RAGTruth's refusal sentence, leaving 5,593 attempted answers, 1,700 of them with at least one hallucinated span. The three passages are the chunks, scored exactly as the five retrieved chunks are in the main grid. The mea-

	Mean drop			Detected @0.5		Detected @floor gate	
Dataset / generator	**NLI-max**	**E−C**	**AlignScore**	**NLI-max**	**E−C**	**NLI-max**	**E−C**
NQ / Claude Haiku 4.5	0.054	0.067	0.247	5.1%	0.1%	50.2%	51.5%
NQ / Qwen2.5-7B	0.261	0.315	0.436	35.1%	5.8%	71.0%	72.9%
NQ / Mistral-7B	0.251	0.311	0.441	34.3%	8.8%	74.1%	73.9%
NQ / GPT-4o-mini	0.451	0.545	0.640	56.6%	14.0%	85.2%	86.9%
HotpotQA / Claude	0.085	0.110	0.332	13.9%	5.3%	49.7%	51.4%
HotpotQA / Qwen2.5-7B	0.270	0.337	0.494	54.8%	24.6%	69.6%	68.8%
HotpotQA / Mistral-7B	0.287	0.352	0.480	52.5%	26.1%	72.1%	72.6%
HotpotQA / GPT-4o-mini	0.391	0.456	0.661	70.5%	39.1%	83.8%	84.1%

Table 7: The entailment model scored as entailment minus contradiction (E−C), beside NLI-max and AlignScore on the same cases. E−C's mean drop is given on its native $[-1, 1]$ scale, twice the drop on the rescaled one, so it is the most favourable reading of the new scorer. Detection columns use the rescaled score. NLI-max and AlignScore values are those of Table 6.

	Detected @0.5					Mean drop	Untouched pass @0.5	
Cell	**NLI-max**	**Claim**	**Align**	**MC whole**	**MC sent.**	**MC whole**	**MC whole**	**MC sent.**
NQ / Claude	5.1%	25.1%	24.1%	82.7%	98.9%	0.585	91%	68%
NQ / Qwen	35.1%	48.1%	51.8%	95.9%	97.1%	0.626	83%	81%
NQ / Mistral	34.3%	48.2%	51.8%	96.4%	97.8%	0.628	84%	79%
NQ / GPT-4o-mini	56.6%	61.0%	75.4%	97.5%	97.7%	0.765	94%	94%
NQ / Qwen verbose	10.4%	28.9%	16.0%	65.2%	98.8%	0.439	88%	32%
HotpotQA / Claude	13.9%	34.3%	48.9%	80.6%	97.9%	0.524	86%	60%
HotpotQA / Qwen	54.8%	66.8%	74.5%	91.0%	95.1%	0.587	82%	72%
HotpotQA / Mistral	52.5%	59.6%	71.5%	90.5%	94.2%	0.581	81%	64%
HotpotQA / GPT-4o-mini	70.5%	70.6%	89.4%	94.8%	95.2%	0.679	88%	85%
HotpotQA / Qwen verbose	17.4%	43.6%	18.5%	51.0%	97.6%	0.349	88%	16%

Table 8: Bespoke-MiniCheck-7B (MC) on the cases of Table 6 and of the verbosity arm, beside the three evaluators of the main text (NLI-max, claim-min, AlignScore), pooled over the four matched systems. "Whole" scores the full answer as one claim; "sent." takes the minimum over its sentences, as the reference implementation does. Detection is the share of answers passing the gate untouched that fall below it once falsified, so it is computed on the answers in the last two columns. The first three columns are those of Tables 6 and 2.

sures below were fixed before any score was read. The answers are longer than any generator's in our grid, with a median of six assertions.

A RAGTruth answer is either clean or hallucinated, never both, so there is no paired drop to report. We report three measures instead. AUROC is the probability that a clean answer outscores a hallucinated one, with a bootstrap interval over answers. Detection is the share of hallucinated answers below the 0.5 gate. False rejection is the share of clean answers below the same gate.

The scorers rank as they do on the probe. By AUROC, NLI-max separates hallucinated from clean answers least well and MiniCheck best, with claim-min and AlignScore between, and the intervals of neighbouring scorers do not overlap except for MiniCheck's two readings. Within each generator NLI-max is lowest or tied for lowest in all six (Table 10). The 0.5 gate repeats the trade of Section 5.3. The two minimum aggregates catch the most hallucinations, 91.7% for claim-min and 93.9% per sentence, and reject 62.9% and 55.1% of clean answers to do it. AlignScore rejects the fewest clean answers, 7.8%, and catches 36.7%.

A small hallucination is caught least often, under every scorer. Among hallucinated answers, detection rises with the share of the answer the hallucination occupies under all five scorers, with no overlap between the intervals of the lowest and highest thirds in any. AlignScore goes from 18.0% [15.0, 21.4] to 62.4% [58.4, 66.3], NLI-max from 25.0% to 41.3%, and MiniCheck whole from 75.5% to 91.5%. This is the dilution the probe isolates, observed on errors no one constructed. It is not answer length under another name: among hallucinated answers the share and the assertion count are almost uncorrelated (Spearman $\rho = -0.07$).

				Det., by hall. share			AUROC, by assertions			Probe AUROC	
Scorer	**AUROC**	**Det.**	**FR**	**Low**	**Mid**	**High**	**≤5**	**6–7**	**≥8**	**NQ**	**HotpotQA**
NLI-max	.654 [.639,.669]	32.9%	18.6%	25.0%	32.5%	41.3%	.669	.667	.636	.749	.671
Claim-min	.735 [.721,.749]	91.7%	62.9%	87.3%	92.2%	95.6%	.718	.726	.709	.740	.660
AlignScore	.795 [.783,.808]	36.7%	7.8%	18.0%	29.7%	62.4%	.784	.786	.764	.903	.888
MC whole	.837 [.825,.847]	84.2%	33.4%	75.5%	85.7%	91.5%	.831	.833	.803	.968	.947
MC sent.	.823 [.811,.835]	93.9%	55.1%	91.0%	94.3%	96.3%	.820	.816	.773	.967	.947

Table 9: The scorers of Table 8 on 5,593 attempted RAGTruth QA answers (3,893 clean, 1,700 hallucinated), at the 0.5 gate; AUROC with its 95% bootstrap interval, detection (Det.) and false rejection of clean answers (FR). Hallucinated share is the fraction of an answer's characters inside an annotated span, split into thirds of the hallucinated answers (cuts at 0.109 and 0.251; 567, 566 and 567 answers). Assertion bins are the tertile cuts of the whole population; ties at whole numbers leave 2,631, 1,240 and 1,722 answers in them. The last two columns are not from RAGTruth: they are the same scorer's AUROC between untouched and falsified answers over all cases of Table 6, per dataset, computed after the RAGTruth measures as a point of comparison.

Generator	**H / C**	**NLI**	**Claim**	**Align**	**MC**	**MC s.**
GPT-3.5	75 / 808	.640	.733	.721	.738	.721
GPT-4	42 / 886	.544	.545	.544	.602	.632
Llama-2-7B	497 / 474	.538	.602	.705	.712	.739
Llama-2-13B	394 / 586	.558	.665	.680	.776	.739
Llama-2-70B	314 / 605	.575	.694	.735	.821	.762
Mistral-7B	378 / 534	.649	.732	.796	.879	.826

Table 10: AUROC within each RAGTruth generator; H / C is hallucinated / clean answers. GPT-4's 42 hallucinated answers give intervals of about ±0.09.

Length lowers separation. AUROC is lowest in the longest third for every scorer. Relative to the shortest third it falls by 0.03 for NLI-max and MiniCheck whole, 0.02 for AlignScore, 0.05 per sentence and 0.01 for claim-min, whose intervals overlap across thirds. As answers lengthen, NLI-max catches fewer hallucinations, 41.4% to 26.4%. The two minimum aggregates catch more but reject clean answers faster still, claim-min 54.3% to 77.4% and per sentence 39.4% to 83.5%, which is the loss of thresholdability that Section 5.3 reports. Llama-2-7B, the most verbose generator at 8.4 assertions per answer, has the highest false rejection under every scorer.

Natural errors are harder for the stronger scorers. Against their AUROC on our falsified answers, AlignScore and both MiniCheck readings lose 0.09 to 0.14 on RAGTruth. NLI-max sits just below its probe range and claim-min inside it, both being weak on constructed errors already. This is the direction Sinha (2025) report and much less than the extreme they find. The comparison crosses datasets, generators and error types at once, so we read it as a direction and do not size it.

What this does not test is the generator dependence itself. RAGTruth's generators are not ours, and its answers are longer than any in our grid, so Table 10 cannot replicate the ordering of Table 6. It shows the mechanism of that ordering acting on natural errors.

F Response to falsification by assertion count

G Correlation between retrieval quality and faithfulness

Spearman correlation between NDCG@5 and mean faithfulness over the four retrieval systems, in every generator-evaluator combination. With four systems these coefficients carry no significance individually; the pattern across them is what Appendix W reads. The one non-canonical value, +0.63, arises from a tie between two systems.

H Equivalence across margins

Two one-sided tests on per-query faithfulness, over the six pairs of retrieval systems in each cell, across five margins. A claim of equivalence is a claim about one column of this table and one row of it.

I Verbatim overlap and the response to falsification

Overlap is the fraction of answer word 5-grams occurring in the retrieved context as a contiguous token sequence, on tokens with punctuation mapped to spaces, so it is unaffected by the markdown normalisation of Appendix V. Answers under five tokens are excluded rather than scored zero, since one generator writes far more of them

	Claude Haiku 4.5			Qwen2.5-7B			Mistral-7B			GPT-4o-mini		
Assertions	**n**	**NLI-max**	**Claim-min**	**n**	**NLI-max**	**Claim-min**	**n**	**NLI-max**	**Claim-min**	**n**	**NLI-max**	**Claim-min**
1	91	+0.102	+0.102	413	+0.409	+0.409	422	+0.385	+0.385	680	+0.485	+0.485
2	300	+0.076	+0.151	205	+0.152	+0.219	201	+0.135	+0.246	77	+0.360	+0.467
3	194	+0.032	+0.229	103	+0.051	+0.150	97	+0.077	+0.080	26	+0.041	+0.401
4	103	+0.024	+0.162	40	+0.024	+0.076	50	+0.031	+0.075	10	+0.204	+0.192
5 or more	112	+0.024	+0.146	39	+0.072	+0.126	30	+0.082	+0.139	7	+0.032	+0.002

Table 11: Drop in score when one grounded value is falsified, by number of assertions in the answer, on Natural Questions. Generators are ordered by assertions per answer, most first. Under NLI-max the response declines as answers accumulate assertions in all four generators. Claim-level aggregation removes the decline on Claude alone, where the response at three assertions is larger than at one; on the other three it softens the decline without removing it, and on Mistral-7B at three and four assertions the two aggregates nearly coincide. The thin buckets should be read as such: the GPT-4o-mini rows beyond two assertions rest on 26, 10 and 7 cases, and the last Mistral-7B row on 30. In the single-assertion row the two aggregates coincide exactly, which is the degenerate case they must reduce to.

Generator	**Evaluator**	**Natural Questions**	**HotpotQA**
Claude	NLI-max	-1.00	+0.20
	Claim-min	-0.60	+0.60
	AlignScore	-0.20	+1.00
Qwen	NLI-max	-0.80	+0.60
	Claim-min	+0.63	-0.40
	AlignScore	-0.40	+1.00
GPT-4o-mini	NLI-max	-0.80	+0.80
	Claim-min	-0.80	+0.80
	AlignScore	0.00	+1.00
Mistral-7B	NLI-max	+0.80	+0.60
	Claim-min	+0.80	0.00
	AlignScore	0.00	+0.60

Table 12: With four systems these coefficients cannot carry significance individually. Under AlignScore the pattern is clean in all eight combinations: every HotpotQA coefficient is positive and every Natural Questions coefficient is zero or negative. Under NLI-max it holds in seven of eight, the exception being Mistral-7B on Natural Questions at +0.80. Claim-min gives the same coefficient there from a different ranking: against the NDCG@5 order, NLI-max swaps the top two systems and claim-min the bottom two. Under claim-min it fails in four of eight, and all four failures are on the two open-weight generators. The one non-canonical value, +0.63, arises from a tie between BGE-M3 and text-embedding-3-small, whose claim-min means agree to four decimals (0.7573); at full float precision the tie breaks and the coefficient is +0.80.

and scoring them zero would manufacture the difference the control tests for. Grouping the 6,133 cases that remain into four overlap bands leaves the mean response to falsification at +0.052 to +0.085 for Claude, +0.202 to +0.364 for Qwen, +0.239 to +0.299 for Mistral-7B and +0.326 to +0.507 for GPT-4o-mini. Claude is lowest and GPT-4o-mini highest in all four bands, including the band above 0.3 where all four generators reuse a substantial part of the retrieved wording, and Claude is separated from GPT-4o-mini by at least fourfold throughout. The two open-weight generators swap places in that top band only. The 95% confidence intervals on the four correlations are -0.004 to +0.096 (Claude), +0.127 to +0.229 (Qwen), -0.008 to +0.102 (Mistral-7B) and +0.118 to +0.212 (GPT-4o-mini).

J Retrieval success against correctness

K Checking the judge against a human annotator

Every correctness figure in the paper comes from one judge, and so does the refusal classification behind the correct-versus-incorrect comparison of Appendix U and Appendix J. The judge is Claude Opus 5 at low effort, with the model's default reasoning left on. It is shown the question, the reference answers and the answer, and asked for one word. The answer is CORRECT if it conveys the reference, even phrased differently, with extra correct detail or inside a longer sentence. It is INCORRECT if it contradicts the reference, states a different value or omits it. It is REFUSAL if it declines or says the information is not available. Temperature cannot be set on this model, so the one-word output is the only constraint on its variation.

To check the judge, we drew 25 judged answers from each of the eight dataset-generator cells of the matched grid, pooled over the four embedding systems, with a fixed seed: 200 answers. The author labelled them with the judge's three labels and its definitions, verbatim. The annotator saw what the judge saw, namely the question, the reference answers and the answer, under an opaque identifier and in shuffled order. The judge's verdict, the generator, the system and the retrieved context

Dataset / generator	Evaluator	±0.01	±0.02	±0.03	±0.05	±0.10
NQ / Claude	NLI-max	0/6	4/6	6/6	6/6	6/6
	Claim-min	0/6	0/6	2/6	6/6	6/6
	AlignScore	3/6	6/6	6/6	6/6	6/6
NQ / Qwen	NLI-max	0/6	3/6	6/6	6/6	6/6
	Claim-min	0/6	0/6	6/6	6/6	6/6
	AlignScore	1/6	6/6	6/6	6/6	6/6
NQ / Mistral-7B	NLI-max	0/6	0/6	4/6	6/6	6/6
	Claim-min	0/6	0/6	2/6	6/6	6/6
	AlignScore	0/6	6/6	6/6	6/6	6/6
NQ / GPT-4o-mini	NLI-max	0/6	3/6	6/6	6/6	6/6
	Claim-min	0/6	3/6	6/6	6/6	6/6
	AlignScore	1/6	6/6	6/6	6/6	6/6
HotpotQA / Claude	NLI-max	0/6	0/6	0/6	5/6	6/6
	Claim-min	0/6	0/6	0/6	3/6	6/6
	AlignScore	0/6	2/6	5/6	6/6	6/6
HotpotQA / Qwen	NLI-max	0/6	0/6	0/6	6/6	6/6
	Claim-min	0/6	0/6	0/6	6/6	6/6
	AlignScore	0/6	0/6	0/6	4/6	6/6
HotpotQA / Mistral-7B	NLI-max	0/6	0/6	0/6	0/6	5/6
	Claim-min	0/6	0/6	0/6	3/6	5/6
	AlignScore	0/6	0/6	1/6	5/6	6/6
HotpotQA / GPT-4o-mini	NLI-max	0/6	0/6	1/6	4/6	6/6
	Claim-min	0/6	0/6	1/6	4/6	6/6
	AlignScore	0/6	0/6	0/6	4/6	6/6

Table 13: System pairs judged equivalent, of six tested, across margins, with generators ordered by assertions per answer. A claim of equivalence is a claim about a particular column — and about a particular row: at ±0.03 on NQ/Claude, NLI-max calls all six pairs equivalent and claim-min two, on the same answers. The two rows where ±0.10 still leaves a pair non-equivalent are both HotpotQA under Mistral-7B, the cell with the largest spread in Table 4.

Generator	Dataset	Share of queries	Of that cell, refusals	Attempted and wrong
Claude	NQ	17.4%	69.2%	5.4%
	HotpotQA	34.0%	86.8%	4.5%
Qwen	NQ	22.9%	50.2%	11.4%
	HotpotQA	40.6%	73.8%	10.6%
GPT-4o-mini	NQ	31.7%	79.0%	6.7%
	HotpotQA	36.3%	78.1%	7.9%
Mistral-7B	NQ	26.3%	57.9%	11.1%
	HotpotQA	48.4%	79.7%	9.9%

Table 14: Correctness from graded judgement, over the thirty-two judged cells of the matched grid. The raw cell runs from 17 to 48 per cent of queries; the share that is a grounded, committed, wrong answer runs from 4.5 to 11.4 per cent. The two open-weight generators refuse least often within this cell and therefore leave the largest committed-wrong share, so the cell's size and its interpretable remainder do not order the generators the same way.

	Judge			
Human	Correct	Incorrect	Refusal	Total
Correct	120	1	0	121
Incorrect	0	18	0	18
Refusal	1	1	59	61
Total	121	20	59	200

Table 15: The judge against a blind human annotator on 200 answers, 25 from each dataset-generator cell of the matched grid. Agreement is 98.5% over the three labels ($\kappa = 0.97$) and 99.0% both for correct versus not and for refusal versus not ($\kappa = 0.98$ each).

were hidden, and the annotator did not see the verdicts until labelling was complete. The measures were fixed before any label was collected: agreement with a Wilson interval and Cohen's κ over the three labels, over correct versus not, which is what every accuracy figure uses, and over refusal versus not, which is what the correctness analyses take from the judge.

Table 15 gives the result. Agreement is 98.5% over the three labels, 197 of 200 (interval 95.7 to 99.5%, $\kappa = 0.97$). It is 99.0% both for correct versus not ($\kappa = 0.98$) and for refusal versus not ($\kappa = 0.98$), each with interval 96.4 to 99.7%. The two graders mark the same share correct, 60.5%, because the two answers they disagree on in that respect err in opposite directions. Agreement is 98% for Claude, Qwen and Mistral-7B and 100% for GPT-4o-mini, on 50 answers each, and 99% and 98% on Natural Questions and HotpotQA,

on 100 each. No generator or dataset stands out, though at 50 answers per generator the intervals are wide.

The annotator had marked all three disagreements as borderline while labelling, each time naming the judge's label as the alternative. Two of them decline and then say something anyway, the kind of answer Appendix U places at the boundary between refusing and attempting. A Mistral-7B answer calls the answer unstated and then offers "could be Cessnock" (human refusal, judge incorrect). A Claude answer names the program the reference gives and then declines the question as asked (human refusal, judge correct). The third is a Qwen answer to a HotpotQA question that runs two people together. It separates them and names the reference as the director (human correct, judge incorrect). These show that a judge and a human can read such answers differently. They do not show that either reading is wrong.

The check has three limits. It has one annotator, who is also the author, so it bounds how far the judge departs from one careful reader and not from human judgement in general; a second annotator would show how far two people agree. Incorrect is the thinnest class, with 18 human and 20 judge labels, so agreement on it is the least precise. And the sample contains none of the 220 decline-then-answer rows that our rule calls refusals and the judge grades correct (Appendix U). At 2.7% of the rule's refusals, one or two were expected among the 54 the rule finds in the sample.

Two further observations were made after the labels were read, and were not part of the design. First, the string refusal rule that sets the population for every evaluator score (Section 4) agrees with the annotator on refusal versus not in 96.5% of answers (interval 93.0 to 98.3%, $\kappa = 0.91$), against 99.0% for the judge. It never calls an attempted answer a refusal, and it misses 7 of the 61 answers the annotator calls refusals. Four of the seven give part of an answer and then decline the rest of the question, two say the answer is not given and then guess, and one is a plain refusal worded "I cannot definitively answer", which the phrase list does not cover. The rule therefore leaves some refusals in the attempted population, in line with the judge finding up to 3.4 more refusals per hundred answers (Appendix T). Second, correctness here means agreement with the dataset's reference. In 8 of the 18 answers the annotator graded incorrect, from 7 distinct questions, the annotator's note records that the answer looked defensible: the reference appeared wrong or out of date, or the answer was another valid reading of an ambiguous question. Both graders grade against the reference, as the definitions require, so this does not affect agreement. It does mean that the incorrect side of the correct-versus-incorrect comparison includes some answers a reader might accept. One annotator's notes do not measure how large that share is.

L What the comparison could have detected

Five of the eight cells in Table 4 are null under every evaluator, and Appendix X shows the procedure separates systems that genuinely differ. Neither of those says what size of effect it would have caught. The MDE column of Table 4 gives the minimum detectable effect at 80% power for a two-sided paired test at $\alpha = 0.05$, computed from the observed standard deviation of the paired per-query differences between the best and worst system in each cell.

Two things follow, and the first is a caution about our own null results. The minimum detectable effect runs from 0.012 to 0.064 with a median of 0.036, and in twenty of the twenty-four dataset-generator-evaluator cells it exceeds the spread actually observed. Where we report no detectable difference, an effect the size of the one we measured would usually not have been detected. The nulls bound what a practitioner would notice at this sample size; they are not evidence that the systems are identical, which is why Section 8 tests equivalence separately instead of reading the nulls as equivalence. All four comparisons that do reach significance have a spread above their own minimum detectable effect (0.0461 against 0.0368, 0.0406 against 0.0350, 0.0796 against 0.0614 and 0.0699 against 0.0638), and no other comparison does.

The second is about the evaluators, and it qualifies how Section 7 should be read. Among the three evaluators of Table 4, AlignScore has the lowest minimum detectable effect in all eight cells, 0.012 to 0.037, against 0.021 to 0.061 for NLI-max and 0.024 to 0.064 for claim-min. The generative fact-checker scored over the same answers is lower still in six of the eight, at 0.011 to 0.034 (Appendix N). Some of that is a variance property: the standard deviation of the paired per-query dif-

ference is 0.12 to 0.30 under AlignScore and 0.20 to 0.48 under NLI-max, on identical answers. So part of why the verdict changes with the evaluator is that the evaluators differ in resolving power at fixed n, and not only in how they respond to a known error. Appendix L.1 separates the two by dividing each spread by its own paired standard deviation. AlignScore's standardised effect remains the larger in five of eight cells, so the variance difference does not account for all of it, although it reverses on Mistral-7B.

L.1 Separating resolving power from response

The minimum detectable effects above confound two things. AlignScore may break cells because it responds more to whatever differs between the systems, or because its per-query differences are quieter and the same difference clears significance sooner. The first is a claim about what the instrument resolves; the second is a claim about its variance. Both would produce the pattern in Table 4.

Dividing each cell's spread by the standard deviation of its own paired per-query differences removes the variance term and leaves a standardised effect, comparable across evaluators that do not share a scale. It is the Std. column of Table 4.

AlignScore's standardised effect exceeds the entailment aggregate's in five of eight cells, by factors of 1.3 to 5.6. So the difference between these evaluators is not only that one is quieter. It is not a fixed property of the evaluator either: on Mistral-7B the entailment aggregate has the larger standardised effect in both datasets, and the largest single value in the column is its 0.177 on HotpotQA/Mistral-7B. Claim-min, shown for completeness, is the largest of the three in two cells, HotpotQA/Claude (0.085) and NQ/GPT-4o-mini (0.071). In the other five cells, on the same answers, AlignScore resolves a larger difference between the retrieval systems relative to its own noise. Since the variance has been divided out, that is a statement about what it measures.

Two qualifications. First, the exceptions: besides the two Mistral-7B cells, NQ/GPT-4o-mini reverses the ordering with the two nearly indistinguishable (0.062 against 0.066), in the cell with the tersest generator on the dataset with the narrower retrieval spread. Second, this analysis separates variance from response without identifying what AlignScore is responding to. It is consistent with AlignScore tracking a property of the retrieved context that the entailment aggregate discards when it takes a maximum over chunks, and we do not test that.

M An omnibus test, and the system by evaluator interaction

The spread in Table 4 is a maximum minus a minimum over four systems, so the two systems entering it are chosen for being extreme and its p-value is not adjusted for that choice. Two further tests remove that objection, and they qualify the headline.

An omnibus that re-selects the extremes. Permuting system labels independently within each query and recomputing the range makes the selection part of the null, since the extremes are chosen afresh in every resample. Over 10,000 resamples, and Holm-corrected across the twenty-four comparisons, **two comparisons survive** rather than four: HotpotQA/GPT-4o-mini under AlignScore at permutation $p = 0.0015$, and HotpotQA/Mistral-7B under NLI-max at $p = 0.0019$. HotpotQA/Mistral-7B under claim-min ($p = 0.0070$) and HotpotQA/Qwen under AlignScore ($p = 0.0076$) do not survive. A Friedman test over the same four systems asks the same question by ranks, and leaves **one** significant after the same correction: HotpotQA/Mistral-7B under NLI-max at $p = 0.0005$. Its next smallest uncorrected values are 0.0032 on that same cell under claim-min and 0.0092 on a cell the paired test does not pick out. Both omnibus tests are stricter than the spread test and both now retain a cell, where at eighteen comparisons the permutation test retained one and Friedman none.

So the count is four under a paired test on the extreme pair, two under a permutation test that pays for selecting that pair, and one under a rank-based omnibus. The paired figure is the test a practitioner comparing their adopted model against the best alternative would run. It is also the most permissive of the three, so a reader should see all of them.

Do the evaluators disagree about the systems? A verdict changing is not the same as two instruments ranking four systems differently. We test the second directly. Scores are standardised within evaluator, since the three share neither scale nor variance, and fitted as `z ~ system * evaluator` with a random intercept per query, by maximum likelihood so that the fixed-effects likelihood ratio is valid.

Dataset	Generator	n	LR χ^2_6	p
NQ	Claude	9,408	6.57	0.36
	Qwen	9,480	1.63	0.95
	GPT-4o-mini	7,536	2.42	0.88
	Mistral-7B	8,304	1.41	0.96
HotpotQA	Claude	6,204	3.18	0.79
	Qwen	5,988	7.08	0.31
	GPT-4o-mini	6,120	3.77	0.71
	Mistral-7B	5,076	1.47	0.96

Table 16: Interaction between embedding system and evaluator, on scores standardised within evaluator, with a random intercept per query. A Wald test on the same six terms agrees to two decimal places in every row. No cell shows a detectable interaction.

The interaction is undetectable in all eight cells, p from 0.31 to 0.96, and it is weakest of all on HotpotQA/Mistral-7B ($p = 0.96$), the cell with the largest spread in Table 4. Appendix L.1 puts AlignScore's standardised effect ahead in five of eight cells. Taken together, the three evaluators appear to see much the same thing about these four systems and to resolve it with different clarity. A study using the common entailment aggregate then reports no difference and a study using AlignScore reports one, and both are reading the same underlying pattern through instruments of different resolving power. Refitted with the fact-checker as a fourth evaluator (Appendix N), one cell falls below 0.05 for the first time, HotpotQA/Mistral-7B at $p = 0.041$. It is one of eight tests and does not survive correction over them.

That is a narrower claim than evaluator disagreement and it has the same practical consequence: the published verdict depends on the evaluator. It is also the more defensible one at this sample size, and we state it in preference to the stronger reading. What we cannot do is rule out disagreement, since an interaction test on four systems at these spreads inherits the resolution limits of Appendix L; absence of a detectable interaction is not evidence that none exists.

N A fourth evaluator in the system comparison

Two of the three evaluators of Table 4 share an entailment model. To test the comparison beyond it, we scored all 32,000 answers of the grid with Bespoke-MiniCheck-7B on the whole answer, with the configuration and inputs of Appendix D. Re-scoring the first three probe answers of each of the thirty-two dataset-generator-system cells reproduced the probe's stored scores exactly. The comparison is the one every evaluator gets: attempted answers by the string rule, paired on the queries all four systems answered, the spread between the best and worst system, and a paired bootstrap. The reporting rule was fixed before any score existed. The twenty-four comparisons of Table 4 remain the design family, and the thirty-two comparisons of four evaluators are reported beside them as an extension, Holm-corrected over the thirty-two.

Over the thirty-two, seven comparisons survive, in four of the eight cells (Table 17). The four of Table 4 all survive the larger family, at corrected p of 0.0090, 0.0145, 0.0308 and 0.0468. The fact-checker adds three, all on HotpotQA: Claude, GPT-4o-mini and Mistral-7B. HotpotQA/Claude is new, since none of the other three evaluators registers a difference there. All four Natural Questions cells stay null under every evaluator. The two omnibus tests of Appendix M, extended in the same way, each keep three of the thirty-two. Friedman keeps the fact-checker on HotpotQA/GPT-4o-mini and HotpotQA/Mistral-7B and NLI-max on HotpotQA/Mistral-7B. The permutation test keeps the same two fact-checker comparisons and AlignScore on HotpotQA/GPT-4o-mini.

The fact-checker resolves finer than the other three. Its minimum detectable effect is 0.011 to 0.034, the lowest of the four in six of the eight cells, although on NQ/Claude by less than 0.0002; AlignScore's is lower on NQ/Qwen and NQ/Mistral-7B. It is still above the observed spread in five of the eight. Its standardised effect is the largest of the four in five cells. So the resolving-power reading of Section 7 carries over: the instrument with the finest resolution in most cells registers the most differences.

Refitting the interaction model with four evaluators gives p from 0.46 to 1.00 in seven cells and 0.041 on HotpotQA/Mistral-7B, the cell with the largest spreads in Table 4. That is the first cell in the paper below 0.05. It is one of eight tests and does not survive correction over them (Holm 0.33), so we do not read it as evidence that the evaluators rank the systems differently. It is where such evidence would first appear.

Two further results come from the same scores. The sentence-minimum aggregate, reported in no family, has an uncorrected p below 0.05 in six of the eight cells, the two exceptions being NQ/Claude and NQ/Mistral-7B. And under the

	Natural Questions					HotpotQA				
Generator	Spread	MDE	Std.	p_{32}	Int. p	Spread	MDE	Std.	p_{32}	Int. p
Claude Haiku 4.5	0.0080	0.012	0.067	1.0000	0.464	**0.0272**	0.024	0.141	**0.0405**	0.835
Qwen2.5-7B	0.0105	0.019	0.056	1.0000	0.980	0.0253	0.032	0.098	0.6644	0.603
Mistral-7B	0.0056	0.020	0.030	1.0000	0.998	**0.0501**	0.034	0.201	**0.0031**	0.041
GPT-4o-mini	0.0083	0.011	0.082	0.8148	0.885	**0.0338**	0.024	0.174	<**0.0001**	0.798

Table 17: Bespoke-MiniCheck-7B as a fourth evaluator of the system comparison, on the attempted answers and paired queries of Table 4. Spread, MDE and Std. are defined as there. p_{32} is Holm-corrected over the thirty-two comparisons of the four evaluators, and bold marks the three that survive. Int. p is the system by evaluator interaction of Table 16, refitted with four evaluators.

fact-checker refusals score below attempted answers for all four generators, by 0.44 to 0.71, so like AlignScore it penalises a refusal rather than rewarding it (Table 22).

O The headline count under each refusal rule

Every result from Section 5 onward is computed on the answers a rule calls attempts, and that rule is a researcher degree of freedom. Since this paper's argument is that such choices move conclusions, the count is reported under each defensible rule rather than only under the one we adopted (Table 18).

Three rules are available. The adopted rule strips a leading "Based on . . . ," preamble and then tests the opening against a list of declining phrases wider than the prompt's own refusal sentence. The rule before that change is the same anchored test on the prompt's phrases alone, without the stripping. It grades 759 refusals as attempts: 107 of Claude's, 50 of Qwen's and 602 of Mistral-7B's. Almost all of the difference comes from the wider list rather than from the stripping, since Mistral-7B and Qwen decline in their own words ("The provided context does not . . . ") and seldom with a preamble. The third hands the decision to the judge model used for correctness.

All three rules give the same four comparisons in three of the eight cells, and they agree on which. No comparison breaks under one rule and not another. The Limitations note that swapping detectors carries an *uncorrected* p-value across the conventional threshold in two of the twenty-four comparisons still holds, and the two statements are compatible: the uncorrected values move, and the corrected count does not. That the headline survives all three is a result about this grid and not a general guarantee, since two of the three rules are variations on anchored phrase matching and the third is a single judge model, and a rule built on different principles might not agree.

Rule	**HotpotQA**	spread	p_{holm}
adopted: wider phrase list, preamble stripped			
	Qwen/AlignScore	0.0406	0.0242
	GPT/AlignScore	0.0461	0.0115
	Mistral/NLI-max	0.0796	0.0072
	Mistral/Claim-min	0.0699	0.0378
bare: the prompt's phrases, no stripping			
	Qwen/AlignScore	0.0413	0.0242
	GPT/AlignScore	0.0461	0.0120
	Mistral/NLI-max	0.0686	0.0315
	Mistral/Claim-min	0.0716	0.0161
judge decides the refusal			
	Qwen/AlignScore	0.0433	0.0138
	GPT/AlignScore	0.0432	0.0154
	Mistral/NLI-max	0.0895	<0.003
	Mistral/Claim-min	0.0728	0.0336

Table 18: Comparisons breaking at $\alpha = 0.05$ after Holm correction over the twenty-four, under each refusal rule. All three rules select the same four comparisons and no other reaches significance under any of them. The adopted rows are Table 4's. Paired n moves with the rule: 499, 507 and 467 on HotpotQA/Qwen, 510, 510 and 503 on HotpotQA/GPT-4o-mini, and 423, 500 and 404 on HotpotQA/Mistral-7B, so the populations are not identical. Under the judge, no resample of 10,000 was as extreme as the observed HotpotQA/Mistral-7B NLI-max spread, so its adjusted value is bounded by 24 × 1e-4.

P Is the probe's prefix exchangeable?

Cases are the first 200 eligible answers in dataset order rather than a random 200, so the Wilson intervals beside each detection rate assume a sampling scheme we did not use. If detection does not vary with position inside the prefix, the prefix behaves like a random draw from the eligible set on the quantity being reported.

Detection by quartile of position under the entailment aggregate, pooled over the four embedding systems: on Natural Questions 3.2, 3.1, 7.9

and 6.7% for Claude, 40.4, 33.1, 30.7 and 36.2% for Qwen, 63.5, 55.4, 44.0 and 63.5% for GPT-4o-mini and 39.6, 31.9, 34.9 and 31.2% for Mistral-7B; on HotpotQA 12.8, 13.7, 18.4 and 10.9%, then 55.0, 58.3, 44.4 and 61.0%, then 68.8, 71.2, 65.1 and 78.8%, then 52.9, 47.3, 58.1 and 60.0%. A Spearman test of per-case detection against position reaches $\alpha = 0.05$ in one of the eight cells (NQ/Qwen, $\rho = -0.076$, $p = 0.048$), which does not survive correction over the eight tests ($p \approx 0.39$) and corresponds to almost no relationship in any case. The quartile figures fluctuate without a direction.

This checks position only. It does not rule out some other ordering effect in the datasets that happens not to show up as a monotone trend in detection, so we also ran the probe on a random sample.

A seeded random 200. For every system, dataset and generator we built all eligible cases and kept a random 200 of them (`random.Random(42)`), then scored that sample with the four scorers of Table 22. The measures were fixed before any score was read. The random sample shares 4,512 of its 6,179 cases with the first-200 sample and adds 1,667. HotpotQA/Mistral-7B adds none, since it is a census under both schemes, and neither does HotpotQA/GPT-4o-mini on all-mpnet. On the shared cases NLI-max, claim-min and AlignScore reproduce the first-200 scores exactly. MiniCheck differs by at most 0.004, which changes no detection outcome at 0.5 in any of the 4,512 shared cases.

Table 19 sets the two samples side by side. Across the forty dataset, generator and scorer combinations (the four scorers plus MiniCheck's per-sentence aggregate), detection at 0.5 moves by at most 2.4 points, with a median of 0.4. The random rate falls inside the first-200 Wilson interval in 39 of the 40. The exception is NQ/Claude under per-sentence MiniCheck, 98.9% against 96.9%, which is a saturated rate. Every ordering statement the paper reads off the probe comes out the same on both samples. AlignScore has the largest mean drop in all eight cells, and claim-min detects more than NLI-max in all eight. Claude is the least and GPT-4o-mini the most detected in 8 of the 10 dataset-scorer pairs, and the two exceptions are the same on both samples: per-sentence MiniCheck, where every generator sits between 94 and 99%.

We therefore treat the first-200 sample as exchangeable with the eligible set for the detection rates reported, and keep it as the published sample, since every other table is built on it.

Q The markdown normalisation defect

Sizing the defect before the fix, on 300 answers per cell, stripping markdown lowers Claude's whole-answer entailment score by 0.005 to 0.012 on Natural Questions and 0.021 to 0.043 on HotpotQA, moving it by more than 0.05 on 4.7 to 9.7% and 17.7 to 19.3% of answers respectively. The single-assertion identity is the diagnostic: with one claim and no markdown the whole-answer and claim-level aggregates must be numerically equal, and they were. With markdown they differed on every one of the 131 formatted answers then in the grid, by up to 0.729 on a pair of asterisks. After stripping in both places the formatted group behaves like the unformatted one: 7 of the 140 cases then present differ at all, by at most 1.5e-05, which is batch-padding noise. Over the present four-generator grid the check covers 3,249 single-assertion answers, 152 of them formatted, with a largest absolute difference of 2.33e-05.

R The same probe with entities instead of numbers

A numeric substitution is one error type, and the pattern above could be specific to it. The same construction runs on named entities: take a grounded entity in the answer, replace it with a donor entity that the context does not contain, hold everything else fixed. Cases are built once and every evaluator sees the same ones. Table 20 reports them.

The verbosity effect survives the change of error type. Under the entailment aggregate on Natural Questions a falsified entity moves the score by 0.082 for the most verbose generator and 0.403 for the tersest, against 0.054 and 0.451 for numbers. Detection at the fixed gate is 9.1% against 48.8%, against 5.1% and 56.6%. The gap narrows slightly and does not close.

Ordering by generator holds in three of the six dataset-evaluator combinations, against five of six when only three generators were compared. Both failures the fourth generator introduces are transpositions of Qwen and Mistral-7B under NLI-max, and on the numeric probe those two sit within 0.03 of each other (35.1% against 34.3% detection on Natural Questions, 54.8% against 52.5% on HotpotQA), so little is needed to reorder them. On

			NLI-max		Claim-min		AlignScore		MiniCheck	
Dataset	Generator	Added	First	Random	First	Random	First	Random	First	Random
NQ	Claude	460	5.1	6.3	25.1	26.2	24.1	22.1	82.7	81.0
	Qwen	339	35.1	37.2	48.1	49.2	51.8	52.0	95.9	94.9
	Mistral-7B	196	34.3	34.0	48.2	48.6	51.8	49.8	96.4	96.4
	GPT-4o-mini	81	56.6	56.6	61.0	61.4	75.4	75.1	97.5	97.9
HotpotQA	Claude	371	13.9	15.1	34.3	32.6	48.9	51.3	80.6	81.7
	Qwen	134	54.8	55.4	66.8	66.1	74.5	75.2	91.0	91.3
	Mistral-7B	0	52.5	52.5	59.6	59.6	71.5	71.5	90.5	90.5
	GPT-4o-mini	86	70.5	70.8	70.6	70.9	89.4	88.9	94.8	94.8

Table 19: Falsification detection at the 0.5 gate (%) on the first 200 eligible answers and on a seeded random 200, per dataset and generator, pooled over the four embedding systems. *Added* is the number of cases the random sample contains and the first-200 sample does not; the rest are shared and score identically, up to 0.004 for MiniCheck. MiniCheck is the whole-answer aggregate. HotpotQA/Mistral-7B is a census under both schemes.

the entity probe the Natural Questions swap is not small: Mistral-7B detects 26.9% of falsified entities and Qwen 10.8%. The remaining failure, present with three generators too, is claim-min on Natural Questions, where Claude is scored *more* sensitively than Qwen (0.163 against 0.112, detection 25.9% against 17.9%). The ordering of the extremes survives throughout. Assertion count predicts the ordering well but it does not fix it.

Entity and numeric substitution are not interchangeable, and neither is uniformly the harder case. Both agree closely for GPT-4o-mini and Mistral-7B, whose mean drops under the two error types differ by at most 0.048 and 0.052 across all six cells. They diverge for Claude and Qwen, and in opposite directions. On HotpotQA the entailment aggregate responds roughly twice as strongly to a falsified entity as to a falsified number for the verbose generator (0.178 against 0.085). On Natural Questions the mid-length generator shows the reverse, by a wider factor (0.088 against 0.261). A claim that an evaluator does or does not catch fabrications is therefore a claim about a particular kind of fabrication. The two error types we tested do not license extrapolation to each other. They certainly do not license it for unsupported causal claims or contradicted negations, which we did not test.

S The effect is not verbatim copying

A natural explanation is that one generator quotes the retrieved text back, so the evaluator scores a copy of its own premise and one substituted value cannot move it. The data does not support it. Verbatim overlap with the retrieved context is 0.226 for Claude, 0.194 for Qwen, 0.165 for Mistral-7B and 0.235 for GPT-4o-mini. The explanation needs that overlap to run opposite to sensitivity, and it does not: the generator that copies most is the most sensitive, and the one that copies least sits in the middle. The correlations between overlap and the response to falsification are +0.046, +0.178, +0.046 and +0.166, where the explanation needs all four to be negative. The separation also survives at matched overlap, in every band including the one above 0.3 (Appendix I).

T Membership is set by a grader, and graders disagree

Excluding refusals means deciding which answers are refusals, and a rule decides it. We compare two: the rule every evaluator score in this paper rests on, which matches a list of declining openings wider than the prompt's own refusal sentence after stripping a leading attribution clause (Section 4), and the refusal classification of a judge model shown the question, the reference answers and the answer. They agree on most rows. The judge finds up to 3.4 more refusals per hundred answers in a system, and on one system 0.5 fewer, and it reorders the four systems by refusal rate in none of the eight dataset-generator cells. The comparison below covers all twenty-four dataset-generator-evaluator comparisons, and the graded-correctness results in the next subsection cover all thirty-two dataset-generator-system cells.

A level shift that preserves the ordering is usually unremarkable. This one is not, because the rows it moves sit at the boundary between refusing and attempting. Swapping detectors changes the status of 68 rows on NQ/Claude and 173 on HotpotQA/Claude, and of between 5 and 317 per dataset-generator cell across the grid. In two of

		Claude Haiku 4.5		Qwen2.5-7B		Mistral-7B		GPT-4o-mini	
Dataset	**Evaluator**	Δ	Det.	Δ	Det.	Δ	Det.	Δ	Det.
NQ	NLI-max	0.082	9.1%	0.088	10.8%	0.203	26.9%	0.403	48.8%
	Claim-min	0.163	25.9%	0.112	17.9%	0.233	37.1%	0.431	53.4%
	AlignScore	0.232	21.9%	0.290	33.1%	0.399	45.4%	0.673	77.5%
HotpotQA	NLI-max	0.178	28.6%	0.267	54.1%	0.312	54.6%	0.412	75.1%
	Claim-min	0.132	38.2%	0.262	67.3%	0.266	60.0%	0.412	75.9%
	AlignScore	0.321	47.6%	0.463	69.4%	0.429	64.4%	0.668	90.6%

Table 20: Response to a falsified *entity*, on the same answers and the same fixed 0.5 gate as Table 1. Δ is the mean drop in score. n = 783, 740, 720 and 725 on NQ and 800, 771, 572 and 762 on HotpotQA; detection is computed over the cases scoring above the gate before falsification. With three generators the generator ordering of Table 1 was reproduced in five of six rows. With four it is reproduced in three of six, one of them (NQ under AlignScore) only because Qwen and Mistral-7B tie on the numeric probe at 51.8%. Two of the three failures are transpositions of those two mid-range generators under NLI-max, which sit within 0.03 of each other on the numeric probe. The third is claim-min on NQ, where Claude and Qwen swap, as it did with three generators. Generators are ordered as in Table 1. The ordering of the extremes — Claude lowest, GPT-4o-mini highest — holds in every row under both error types.

the twenty-four comparisons that carries the uncorrected p across 0.05: NQ/Claude under the entailment aggregate from 0.155 to 0.032, and HotpotQA/Claude under AlignScore from 0.063 to 0.022. Neither survives correction. The four comparisons that do survive are the same under both, and under the rule stripped of its two extensions (Appendix O), so we report the corrected count and treat any single comparison near the uncorrected threshold as undetermined.

U The correctness grader admits refusals to the wrong population

The same problem returns once faithfulness is conditioned on correctness. String containment marks an answer correct when a reference string occurs in it, and a short reference string can occur inside a refusal. On a 200-answer calibration ten refusals were marked correct this way, and the judge overturned nine. The tenth declines and then answers anyway: it opens "I cannot answer based on the provided context" and goes on to name the record holder the question asks for, so the judge grades it correct and does not classify it as a refusal. Each such row carries a near-floor score into the correct-answer average and lowers the baseline every correct-versus-incorrect comparison is measured against. Under graded judgement no answer the judge classifies as a refusal is graded correct, in any of the thirty-two judged cells. Decline-then-answer rows like the tenth are where the judge and our rule part: 220 answers the rule calls refusals, 2.7% of them, are graded correct by the judge. This is why the correctness analyses take refusals from the judge rather than from the rule.

That is enough to change a sign. With containment, and a baseline excluding refusals from the incorrect side only, NQ/Claude answers looked more grounded when wrong than when right by 0.018 to 0.050. Excluding refusals from both sides gives -0.006 to +0.018; grading with the judge gives -0.051 to -0.113. Each step corrects a grader, not an answer, and all three move the quantity the same way. Tested directly over the thirty-two judged cells, a permutation test with Holm correction over that whole family finds eight significant, every one of them negative, and none of the five positive-signed cells has an interval excluding zero. Correcting within each generator's own eight tests instead returns nine, so this count depends on the family chosen and we report the stricter one. E5-instruct on Natural Questions is negative under all four generators (-0.105, -0.115, -0.123, -0.125) and survives the family correction under three of them, Claude's falling just outside it at 0.072. We report no evidence that answers are more grounded when wrong. Intervals are ± 0.064 to ± 0.129 on 44 to 134 wrong answered rows per cell, so this bears on the general claim rather than on individual systems.

The same accounting applies to the contingency between retrieval success and correctness (Appendix J). The cell where retrieval succeeded and the answer was still wrong bears on whether retrieval quality is sufficient, and 50 to 87 per cent of it was never attempted. Read whole it runs from 17 to 48 per cent of queries and would support the claim that good retrieval often fails to produce a

correct answer. Most of what it counts is a system declining to answer, and refusal rates differ by more than twenty points across the systems compared. What is left is real, and 4.5 to 11.4 per cent.

V A normalisation defect that produced a result

During development, before the normalisation was fixed, this analysis found two of the four cells then in the design breaking rather than one. The difference was a change of text normalisation, not of test, data or population. The entailment scorer read the answer as the generator wrote it, while the claim-level scorer stripped markdown first, so the two aggregates were not reading the same hypothesis. Markdown is unevenly distributed, appearing in 76.0% of Claude's attempted answers against 1.3%, 0.3% and 0.2% for Qwen, GPT-4o-mini and Mistral-7B, so one generator's formatting habits were entering a comparison between embedding systems. Stripping it for all three evaluators removed the significance of NQ/Claude under AlignScore. That comparison is $p = 0.0081$ uncorrected and 0.154 after correction over the twenty-four in Table 4. None of the four comparisons that now survive is affected, since all are on generators that put markdown in at most 1.3% of their attempted answers.

How it was caught is the transferable part. The defect is invisible in every summary statistic and survives every significance test. What exposed it was an identity the two aggregates must satisfy exactly on single-assertion answers. It failed on all 131 formatted answers then in the grid, by as much as 0.729 (Appendix Q). Metrics that admit such an identity should be checked against it.

W Direction is dataset-dependent

Where a difference exists it does not point the same way. Correlating NDCG@5 with mean faithfulness across the four systems, in all twenty-four dataset-generator-evaluator combinations (Appendix G), splits the two datasets. Under AlignScore the split is exact: all four HotpotQA coefficients are positive, reaching +1.00 in three of them, and all four Natural Questions coefficients are zero or negative. Under NLI-max it holds in seven of the eight. Every HotpotQA coefficient is positive and three of the four Natural Questions coefficients are negative, reaching -1.00 for Claude. Mistral-7B on Natural Questions is the exception, at +0.80. Under claim-min the split fails in four of eight, and all four failures fall on the two open-weight generators. Better retrieval accompanies more faithful answers on the multi-hop dataset and not on the single-hop one. Composing an answer out of several retrieved pieces plausibly makes grounding sensitive to how much of the evidence is present, where a single-fact answer is not. We do not test that. The experiment varies embedding model rather than manipulating retrieval quality, so none of this is an effect of NDCG@5 itself.

Claim-min breaks the split. Four of its eight combinations go the wrong way, and all four are on the two open-weight generators. Qwen inverts the split on both datasets, at +0.63 on Natural Questions and -0.40 on HotpotQA. Mistral-7B is positive on Natural Questions, at +0.80, and flat on HotpotQA. Drop claim-min and fifteen of the remaining sixteen combinations follow the split; drop the two open-weight generators instead and all twelve do. The third evaluator and the fourth generator are what show the direction is not a property of the dataset alone.

X A system the test should separate

Five of the eight cells are null. A reader is entitled to ask whether the test detects anything at all, and the grid answers that without a new experiment.

We ran a fifth embedding model, jina-embeddings-v3 (Sturua et al., 2024), over both datasets and all four generators. On Natural Questions it retrieves badly. Its NDCG@5 is 0.685, where the other four sit between 0.786 and 0.830, and it widens the spread the comparison runs on from 4.4 points to 14.5. Paired TOST on per-query NDCG gives mean differences of 0.102 to 0.145 against the other four, at $p_{\mathrm{TOST}} = 1.0000$ each time. The test cannot place a system further from equivalence than that. HotpotQA is a different story. There jina falls inside the band, and its pair with text-embedding-3-small is the only pair in our grid that is both undifferentiated and TOST-equivalent at ± 0.02 (mean difference 0.0059, $p_{\mathrm{TOST}} = 0.0083$).

Rerun Table 4 over five systems instead of four, holding the test, the resampling and the correction fixed, and the count goes from four comparisons in three of eight cells to eight comparisons in five of eight. It does not reach every cell: Natural Questions under GPT-4o-mini and under Mistral-

7B, and HotpotQA under Claude, still show no detectable difference. What changes is which evaluators participate. Among the matched four, only AlignScore breaks on anything but Mistral-7B. With jina included, the entailment aggregate breaks on Natural Questions under Qwen, and claim-min breaks in three further cells. Admitting one system whose retrieval is genuinely worse is enough to make the common evaluator register differences it otherwise reports as null. That is the premise of the matched design failing, not a better result.

This section asks what happens to systems that retrieve about equally well, and a system retrieving ten points worse ought to come out differently. It does, under all three evaluators. So the four nulls do not come from an instrument that never fires. At these sample sizes the procedure separates systems that differ and declines to separate the matched four.

The control bounds our own framing too. Matched retrieval quality has to be checked rather than assumed from a plausible-looking set of models, and one of the five we ran fails the check on one of the two datasets.

Y Floor, Ceiling and the Arrangement of Context

Everything above compares retrieval systems against each other. That comparison fixes neither endpoint, so we measured both. These conditions ran on the Claude arm only, at $n = 1000$ per dataset.

The floor. Given no retrieved context at all, the generator answers 32.2% of Natural Questions correctly and declines 15.5% of the time. Both benchmarks are built on Wikipedia, so roughly a third of the single-hop result is available without retrieval. HotpotQA behaves differently: 20.9% correct, with a 70.4% refusal rate. Asked to compose across documents it does not have, the model mostly declines. Any claim that retrieval produced a correct answer on Natural Questions sits on top of this floor.

The ceiling. Supplying the answer-bearing chunks and nothing else gives 77.0% on Natural Questions and 86.8% on HotpotQA. On HotpotQA that works as a ceiling, with the best of the four systems reaching 81.0% of it and the worst 60.8%. On Natural Questions three of the four exceed it, at 102.2%, 102.6% and 103.4%.

	NQ		HotpotQA	
condition	acc.	ref.	acc.	ref.
all gold	.790	.127	.877	.073
reversed	.773	.123	.890	.063
shuffled	.790	.123	.880	.070
one gold only	.770	.170	**.517**	**.757**
one gold, first	.753	.137	.530	.693
one gold, middle	.757	.163	.503	.683
one gold, last	.727	.153	.510	.723
distractors only	**.090**	.667	**.150**	.913

Table 21: Accuracy and refusal rate when the gold evidence is rearranged, $n = 300$ per condition, Claude. These are the first 300 queries of each dataset, so "all gold" is the ceiling condition of the text on a subset. Its .790 and .877 differ from the ceiling's 77.0% and 86.8% because of the subset, and the ceiling run restricted to the same 300 queries gives .790 and .877 too. The last four conditions hold context length at five chunks. Order is inert. Losing half a multi-hop chain is not, and it converts into refusals instead of errors.

We report those unclipped, because the explanation is a design choice worth stating. Our oracle supplies answer-bearing chunks and nothing around them. That is less context than top-five retrieval, not more. Questions that need the surrounding material get refused under it, 12.9% of the time on Natural Questions. It therefore ceilings evidence selection and not accuracy. On a single-hop dataset the extra material retrieval drags in pays for the extra noise. Where composition is the difficulty it stops paying.

Arrangement. We vary how the gold evidence is presented, over 300 queries per condition (Table 21).

Order is inert. Reversing or shuffling the evidence moves accuracy by at most 1.7 points on either dataset. Holding length at five chunks and moving a single gold chunk between the first, middle and last position spans 0.727 to 0.757 on Natural Questions and 0.503 to 0.530 on HotpotQA. At $k = 5$ we see no position effect.

Subset is not inert, and it bites only where composition does. Cutting the context to one gold chunk costs Natural Questions two points and HotpotQA thirty-six. The refusal column explains the gap: HotpotQA refusals go from 7.3% to 75.7%. Handed half of a multi-hop chain, the model declines instead of completing it from memory. This is the same refusal machinery that Section 6 shows will distort a faithfulness average if those answers are left in it. Here retrieval quality is what triggers it.

Distractors alone give a second floor, and it sits

below the no-retrieval one: 9.0% and 15.0% correct, against 32.2% and 20.9% with no context at all. The refusal column shows why. Handed only irrelevant context under the retrieval prompt, the model declines 66.7% and 91.3% of the time instead of answering from memory.

Z Geometric Analysis and Re-ranking

One obvious response to any of this is to re-rank retrieved chunks by how well they support an answer. Whether that can work depends on a usable signal existing before the answer does. We tested for one.

Two correlations are relevant and they are not interchangeable. Correlating a chunk's similarity to the query with its entailment of the *gold* answer measures a static property of the embedding space. Correlating it with entailment of the *query* measures what a re-ranker could condition on. We compute both, at $n = 200$ per system and dataset.

Natural Questions gives a weak positive signal on both, and the two track each other: 0.146 to 0.217 for the gold-answer version, 0.131 to 0.226 for the query version, significant at $\alpha = 0.05$ in three of four systems each. HotpotQA gives nothing. The gold-answer correlation there is negative in all four systems (-0.086 to -0.115), and the query correlation sits between 0.037 and 0.077 with no p below 0.27.

Re-ranking behaves accordingly. We take the top twenty for one system, re-rank by entailment of the query at $\lambda = 0.6$, and regenerate. Faithfulness on attempted answers moves $+0.008$ on Natural Questions (95% CI $[-0.007, +0.023]$) and -0.008 on HotpotQA ($[-0.029, +0.012]$). Both intervals contain zero. The refusal rate does move, from 15.4% to 19.0% on Natural Questions. Re-ranking takes away context the generator had been using, and it declines instead of answering from less.

A correlation like the one we find on Natural Questions is the natural argument for faithfulness-aware re-ranking. A correlation of 0.13 to 0.23 is not enough to convert into a gain. On HotpotQA there is no correlation to convert.

AA When Retrieval Collapses

Both benchmarks above are Wikipedia, and both are ones these retrieval systems handle well. QASPER (Dasigi et al., 2021) is neither. We ran the same grid over 819 questions about NLP papers with all five embedding systems, and retrieval collapsed: NDCG@5 between 0.062 and 0.122, against 0.705 to 0.830 on the other two datasets, with Recall@5 of 8% to 18%.

The cause is structural. QASPER is single-document question answering. Pooling 819 papers into one corpus of 13,876 chunks leaves a question like "which multilingual approaches do they compare with?" carrying no signal about which paper it belongs to. We would not offer this as a fair test of the retrievers.

Refusals rise, to 27.8%, 35.4% and 45.0% across the three generators run on it, and between 2,254 and 2,955 answers per generator survive. On those answers, two orderings from the main grid invert, and the second of them for one generator only.

The generators swap places. GPT-4o-mini is the most faithful here under all three evaluators and Claude the least. In the main grid claim-min already ranks them that way, but NLI-max puts Claude first on both datasets (0.904 and 0.706, mean over all attempted answers, which is a different population from the probe cases of Table 1), so on that evaluator QASPER reverses the ordering. The evaluators swap as well, and that matters more. Across all eight cells of the main grid the three order identically by mean score over attempted answers, AlignScore above NLI-max above claim-min. The probe cases of Table 1 are a narrower population and need not follow it: there NLI-max sits above AlignScore in one cell, NQ/Claude (0.906 against 0.866). On QASPER that ordering holds for Qwen and, by 0.003, for GPT-4o-mini, and inverts for Claude, which reads NLI-max 0.747, AlignScore 0.625 and claim-min 0.249.

The 0.249 carries the information. Claim-min reports the least supported assertion in an answer, so a value that low says Claude is producing multi-assertion answers whose weakest component the context does not support. That is what we would expect when the context usually lacks the evidence and the generator answers anyway. NLI-max takes a maximum over chunks, finds something to entail, and reports 0.747 on the same answers.

We keep this out of the main grid, and these numbers are not comparable to the ones above. A population where retrieval failed cannot test a claim about systems at matched retrieval quality. The inversion also has a competing explanation we cannot rule out. With Recall@5 between 8% and

18%, scores are compressed toward the bottom of each evaluator's range, and orderings among compressed scores can reverse through floor effects and distribution shift without any change in what the evaluators are sensitive to. Read this as one boundary observation, not as evidence that evaluator orderings are generally unstable across domains. What it establishes is narrower: the ordering held in every cell we measured where retrieval worked, and it stopped holding for one of the three generators in the one regime we measured where retrieval did not.

AB Does this reach past entailment scorers?

Two of the three evaluators we test are built on the same entailment model, which invites the reading that we have diagnosed NLI rather than faithfulness measurement. Other paradigms exist. ARES (Saad-Falcon et al., 2024) fine-tunes lightweight judges on synthetic data and calibrates them against a few hundred human annotations with prediction-powered inference. RAGVUE (Murugaraj et al., 2026) decomposes a RAG system into retrieval quality, answer relevance, claim-level faithfulness and judge calibration, and attaches an explanation to each. RAG-Zeval (Li et al., 2025) trains a compact evaluator with reinforcement learning to judge faithfulness and correctness in one reasoning pass, rewarding it on rankings of responses rather than on absolute scores. We ran none of the three, so what follows is an argument about what our measurements imply for them, not a test of them.

Three of our results do not depend on entailment. AlignScore is not an NLI model. It is an alignment function trained across factual-consistency tasks, and in the verbosity control it loses more than either entailment aggregate: detection falls from 74.5% to 18.5% on HotpotQA, and its mean response to a falsified value drops from 0.494 to 0.127 (Section 5.3). The structural bias therefore attaches to scalar aggregation over a multi-assertion answer, not to the entailment model underneath.

We tested how far that reaches with a generative fact-checker, Bespoke-MiniCheck-7B, scoring the whole answer as one claim (Appendix D). It is a much stronger instrument on this task. It catches 80.6% to 97.5% of falsified values across the four generators, and the distance between the least and most detected generator falls from over 50 points under NLI-max to about 15. The direction survives. Claude is still the least detected generator on both datasets, and the within-model verbosity arm still costs it 31 and 40 points of detection, from 95.9% to 65.2% and from 91.0% to 51.0%. So the level of the bias depends on the scorer, and a trained checker reduces it sharply; the dependence on how much the answer asserts is present in every scorer we ran. The same checker, scored as a fourth evaluator over all 32,000 answers, also reaches the verdict result. It finds differences between the four systems in three HotpotQA cells, one of them null under every other evaluator, and none on Natural Questions (Appendix N).

Claim-level decomposition is the obvious response and is what RAGVUE prescribes. We measure it. Claim-min is the most length-robust of the three. It gives up 36 detection points between the extreme generators, against 54 and 46 for the others, and it holds 43.6% detection where the others fall below 19%. It is also the aggregate whose absolute scores collapse at that verbosity, to means of 0.234 and 0.076, so decomposition buys sensitivity and costs thresholdability. That trade follows from taking a minimum rather than from the model taking it, so we would expect it in any claim-decomposed metric that aggregates by minimum. We tested two. MiniCheck's own per-sentence minimum keeps detection at 94% to 99% in every cell, verbosity arm included, while the share of the verbose arm's untouched answers passing the 0.5 gate falls to 32% and 16%.

Calibration is a different axis from the one we measure, which is why ARES is complementary rather than an answer. Prediction-powered inference corrects a judge's bias against a human-labelled sample: it makes the reported number closer to what annotators would say on that distribution. Our two failure modes survive that correction. A judge can be unbiased on average and still fail to separate two systems at the sample sizes used here, four hundred to eight hundred paired queries per cell. That is resolving power (Appendix L). It can equally be unbiased on average and still move by 0.054 when a grounded value is replaced with a false one. That is sensitivity to a known error. Neither is a calibration error, and neither would be visible in agreement with annotators on unperturbed answers.

What would test this properly is running these frameworks through the same probe. The

MiniCheck result makes a prediction for them: a trained judge will sit at a higher level than an entailment aggregate, and its detection will still fall when the same generator writes more. We expect the calibration machinery to be orthogonal to that, and we have not shown it.

AC Evidence behind the recommendations

This appendix gives the evidence and the caveats for the recommendations of Section 9. Appendix AD turns them into steps for a verbose generator, and Appendix AE sets out the three evaluators' measured properties side by side.

The evaluation population. Where a framework already scores refusals separately, the refusal-pooling result of Table 3 is a caution about aggregate reporting rather than a defect in the framework. It reached our own pipeline during development, where two implementations of the shared refusal rule disagreed on 365 rows. That is the more realistic failure: the rule exists, and two parts of the pipeline apply different versions of it. We found it before computing any number reported here, and every reported number comes from the single implementation that replaced both.

Claim-level aggregation. For comparing systems the relevant property is a different one (Appendix AE). For checking individual answers, claim-level aggregation is a partial remedy. It lifts detection on the verbose generator's NQ cell almost fivefold, from 5.1% to 25.1%, and 25.1% is still a minority of known fabrications, against 61.0% for the tersest generator under the same aggregate. The gap between generators narrows and does not close, so claim-min reduces the structural bias rather than removing it. It also costs something the maximum does not. Its absolute scores collapse at that verbosity, from a mean of 0.693 to 0.234 on Natural Questions and 0.424 to 0.076 on HotpotQA, because a minimum over five assertions sits near zero whatever the answer says. Resisting the structural bias and remaining thresholdable are separate properties, and claim-level aggregation buys the first at the cost of the second: a fixed grounding gate carried across to it unchanged would reject almost every verbose answer. Per-assertion thresholding, or weighting assertions by how much of the answer's content they carry, are the obvious next steps and we did not test them.

The deterministic check. Requiring every value in an answer to appear in the retrieved context flags every falsification we constructed. That part is circular and we state it as such: our second eligibility rule defines the substituted value as absent from the retrieved context, so a containment test must catch it. The informative comparison is the other direction. The three evaluators of the main text reach 89.4% at best and 5.1% at worst on the same cases, and a generative fact-checker 97.5% and 80.6% (Appendix D), so these edits are not hard to catch — they are hard for an entailment aggregate to catch, and the check is indifferent to answer length where the aggregates are not. What the probe does measure about the check is its false-positive rate, 0.8% to 5.2% on untouched answers, and that rate is not circular.

For a document question answering system, the practical reading of Table 1 is that a grounding threshold on a verbose generator catches a minority of falsified values: 5.1% under the common evaluator, and 25.1% under claim-min, the best of the three on that cell. That is the case for the deterministic check, and two qualifications on it matter. Numbers that assert nothing about the world, such as chunk indices and ordered-list markers, are not expected in the retrieved text, and counting them raises that rate on Claude's HotpotQA answers to about 39%. The rest are quantities the answer derives instead of copying, which a literal check also flags, so it complements an entailment score rather than replacing one.

Checking symbolic content against the source is an old idea (Goodrich et al., 2019; Nan et al., 2021; Zhao et al., 2020), and that numbers are a weak point is documented at the level of the representations too (Wallace et al., 2019). We claim no novelty for it. We do claim the measurement that makes the case for it here, against three current evaluators on identical inputs, and note that a practitioner choosing an instrument needs to know which one works. Entailment is separately a loose proxy for attribution: the AIS framework (Rashkin et al., 2023) separates support by an identified source from general plausibility.

AD What to do with a verbose generator

The detection rates above are low enough to raise a practical question the recommendations so far do

not answer: given a generator that writes several assertions per answer, what should a system actually run? Our measurements support four steps, in order of how much they buy.

Put a deterministic check in front, for anything symbolic. That it flags every falsification we constructed is a consequence of how the probe defines them, not a measurement of the check. The measurement is what it costs — 0.8 to 5.2% of untouched answers flagged — and what the learned scorers do on the same cases, which is 5.1% to 89.4% for the three of the main text and 80.6% to 97.5% for the fact-checker of Appendix D. A check indifferent to answer length is worth having where the aggregates are not. It cannot see unsupported prose, so it is a pre-filter and not a gate.

Do not put a fixed threshold on a long answer. The scale floor moves with verbosity. Under the entailment aggregate, an untouched answer scored against an *unrelated* document averages 0.245 for our terse arm on HotpotQA and 0.509 for the verbose one (Section 5.3). A 0.5 gate therefore sits below the unrelated-document baseline for the verbose arm. Anchoring the gate to that floor costs one extra scoring pass over random contexts and makes the number mean the same thing across generators. Filice et al. (2025) observe that many applications already use a faithfulness check to trigger a fallback, and they evaluate that check with precision- and recall-style metrics tied to answer generation. Our measurements add that where such a gate sits relative to the scale floor depends on the generator's verbosity, so its precision and recall have to be measured per generator.

If you control the generator, constrain the answer. This is the largest lever we measured within one model, and it costs no extra computation. One added prompt line moved Qwen from 1.77 to 4.90 assertions and NLI-max detection from 35.1% to 10.4% on Natural Questions and 54.8% to 17.4% on HotpotQA. On that model the length lever outweighs the choice of evaluator: moving the standard arm from NLI-max to AlignScore gains 16.7 and 19.7 points, against the 24.7 and 37.4 that verbosity costs. Two cautions. We varied length only upward, so the gain from shortening an answer is inferred from the loss on lengthening it, not measured. And the same prompt line also moved the refusal rate (Section 5.3), while its effect on correctness was not measured, so it is not free of side effects.

Choose the evaluator against the failure you care about. Table 22 gives the properties. For separating two systems at a few hundred queries the binding constraint is resolving power, where AlignScore needs the smallest effect. For scoring a single verbose answer it is length robustness, where claim-min gives up the least. The caveat is that its absolute scores fall near zero at five assertions, so it resists the bias without remaining thresholdable.

None of this is a general recipe. Every rate above is a property of a particular generator, evaluator and dataset, and the point of the paper is that those properties are not transferable. What does transfer is the procedure: falsify a value in your own outputs, score the floor against random contexts, and read the detection rate you actually have rather than the one the evaluator's benchmark reports.

AE Choosing among the three

The recommendations above say how to report a faithfulness result. They do not say which evaluator to run, and nothing in this paper supports a single answer to that. The three differ on properties that matter different amounts in different applications. Table 22 collects what we measured into the form that decision needs. Every entry comes from a table earlier in the paper; none of it is new evidence. The rows are grouped by what was measured and on which answers, because the four quantities do not share a unit or a population and cannot be traded against each other.

Read it against the deployment. If answers are short and the question is whether a fabricated value gets caught, the deterministic check dominates, though the evaluator still matters: on the tersest generator's Natural Questions answers NLI-max catches 56.6% of falsified values, AlignScore 75.4% and MiniCheck 97.5%. If answers are long, none of the three main-text evaluators catches most fabrications. The gap between the best and worst of them is still the difference between a weak gate and no gate: 25.1% under claim-min and 24.1% under AlignScore, against 5.1% under NLI-max, on the verbose generator's NQ cell. The fact-checker catches 82.7% there, at the cost of a 7B model per answer, and it still loses 31 and 40 points when one model is prompted to write more (Appendix D). If the evaluation population cannot be guaranteed free of refusals, AlignScore is the only one of the three that scores refusals be-

low attempted answers for every generator we ran. Under the other two, refusals score above real answers for Qwen and GPT-4o-mini and far below them for Claude and Mistral-7B, so pooling them moves the average in a generator-dependent direction. If systems are being compared and the expected difference is small, the relevant property is resolving power, where AlignScore needs 0.012 to 0.037 against 0.021 to 0.061 for NLI-max at the same sample size.

Those criteria do not agree, and we are not proposing a weighting for them. The point of stating them together is that a choice is being made whenever a faithfulness number is reported, and it is usually made by default.

	NLI-max	Claim-min	AlignScore	MiniCheck
Falsification probe, 6,179 cases: detection at the fixed 0.5 gate				
Tersest generator (GPT-4o-mini), NQ–HotpotQA	57–71%	61–71%	75–89%	**95–98%**
Most verbose generator (Claude), NQ–HotpotQA	5–14%	25–34%	24–49%	**81–83%**
Difference, mean of the two datasets (points)	54.1	36.1	45.9	**14.5**
RAGTruth, 5,593 attempted answers: separation of hallucinated from clean				
AUROC	.654	.735	.795	**.837**
System comparison, 423–790 paired queries per cell: score units of each evaluator				
Minimum detectable effect at 80% power	.021–.061	.024–.064	.012–.037	**.011–.034**
All 32,000 answers: mean refusal score against mean attempted-answer score				
Generators whose refusals score higher	2 of 4	2 of 4	**0 of 4**	**0 of 4**

Table 22: What we measured about each instrument, arranged for choosing one. Each block is one quantity on one population, and rows in different blocks are not comparable with each other. Bold marks the best value in each row; within the MDE row, the ranges share no unit across evaluators, and the bolding says only that MiniCheck's is the lowest in six of the eight cells and AlignScore's in the other two (Tables 4 and 17). MiniCheck is Bespoke-MiniCheck-7B scoring the whole answer, on the probe (Appendix D), on RAGTruth and over the 32,000 answers of the grid (Appendix N). Detection is against a falsified numeric value (Table 1); the entity probe puts the three main-text evaluators in the same order by detection in seven of its eight dataset-generator cells (Table 20), the exception being Natural Questions under Claude, which also breaks the order under numeric substitution. The probe ran only where retrieval works. Where it fails, the evaluators' ranking by mean score, a different quantity, inverts for one generator (Appendix AA). RAGTruth's six generators are none of ours (Appendix E). A deterministic value check is not in the table because it is not the same kind of instrument. It covers symbolic content only, and its detection on the probe is 100% by construction, since the probe defines the substituted value as absent from the context. Its measured property is a false-positive rate of 0.8–5.2% of untouched answers.